

A Network Theory of Project Management*

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Abstract

A principal staffs projects by capping how much effort each worker can contribute to each project. Workers prefer some projects over others and work more willingly alongside teammates, but find spreading effort across projects costly. Optimal assignments are characterized by two cutoffs per worker on a single ranking of projects: each worker contributes freely to projects before his first cutoff, is capped below what he would like to contribute between the two, and is unassigned after the second. Because the ranking is common to all workers, project teams are nested. Sorting generalists from specialists depends on a core trade-off: strong ties make a worker valuable to one team but expensive to spread across teams. The sorting index is a ratio of two centralities assessing a worker's influence on collaborators' aggregate effort relative to his local influence within a single team. Assignment selects the network through which shocks propagate, and an improvement local to one project can degrade every other project in the organization.

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1 Introduction

Organizations routinely limit members from doing more of something both parties value. A physician holding a career development award from the National Institutes of Health cannot spend more than a quarter of his time with patients.¹ Engineers in product development have a cap on the number of projects they can join at one time.² An associate at a consulting firm is held below full billable utilization on a project he is passionate about and staffed to work on tasks for smaller clients. In all of the above, the organization values the restricted activity, the member is willing to contribute more effort, yet the limit is administered rather than priced outright.³

While economic theory has largely focused on how organizations shape behavior through incentives, compensation is not the margin most organizations adjust day to day. Performance pay is unavailable in most public and nonprofit organizations due to legal or budgetary hurdles.⁴ In addition, many public sector employees are *intrinsically* motivated to work (Prendergast, 2007). Even in corporations with greater contracting freedom, managers have little to no authority over wages. What they do instead is allocate workers to different projects, which has measurable consequences for firm performance as recent empirical work has found (Minni, 2026).⁵ Wages are not a substitute in any case: they push effort up at the project they reward, and the problem of interest here is limiting it.

To illustrate the economics of assignments, consider a consultant whose manager deliberately limits how much he can work on his favorite project. While this frees up his capacity to be assigned other work the manager values more, it comes

¹NIH Career Development K awards stipulate a split of 75% full-time professional effort and 25% clinical work.

²As multi-project work increases, Wheelwright and Clark (1992) argues that worker productivity may drop because they are not able to allocate their time optimally.

³Management tools like Jira, Time Doctor and Clockify allow managers to monitor and enforce these limits.

⁴Under Title 5 of the United States Code, the yearly monetary incentives a government employee receives cannot exceed a specific threshold. Labor unions frequently file lawsuits against the federal government's implementation of pay-for-performance. See the American Federation of Government Employees' June 2025 memorandum opposing the Office of Personnel Management's overhaul of performance reviews.

⁵See Benson and Shaw (2025) for a broader discussion of the literature on how managers create value through job allocation.

with a trade-off. His colleagues, who work better with him on it, are also less inclined to work on that project, pulling down their contributions there as well. This paper studies how managers— through formal limits on worker activities— organize production by deciding who works on what, how much, and with whom.

I develop a model where a principal (she) assigns workers (he) to projects by placing a cap on the effort each worker can exert at each project. Three economic forces dictate a worker’s choice of effort. First, each worker has an intrinsic preference for each project ([Kristof-Brown et al., 2005](#); [Carpenter and Gong, 2016](#)). Second, he enjoys working alongside others, so his marginal benefit of effort at a project rises with the effort his teammates contribute there ([Mas and Moretti, 2009](#); [Chiocchio et al., 2011](#)). Third, effort is costly and costs come from two sources: concentration on a single project and multitasking ([Leroy, 2009](#); [Coviello et al., 2014](#)).

I focus on management within a business unit or division— teams of analysts, engineers or consultants— so a unit of effort contributes the same to a project no matter who supplies it. Collaborative complementarity enters through worker motivation rather than production technology.⁶ The principal values a profile of project qualities through a single increasing and concave function, where a project’s quality is the sum of effort workers contribute to it. Formally, the principal sets effort caps and workers play a Nash equilibrium in the induced effort game. I provide a condition for a unique equilibrium and show that it is without loss for the principal to directly choose efforts that respect workers’ incentive constraints ([Proposition 1](#) and [2](#)).

Staffing decisions operate on two margins. A positive cap *assigns* a worker to a project, and a zero cap keeps him *unassigned*. Conditional on assignment, a cap set strictly below what a worker would freely contribute towards a project *underutilizes* him. He would like to work more, but is prohibited from doing so, leaving his incentive constraint slack. Setting a cap at or above what he would like to contribute *fully utilizes* him. The two margins differ in how they impact a team. Unassigning a worker from a project eliminates him as a source of collaboration, while underutilization keeps him there as one.

The main structural result is that optimal project assignments organize project

⁶This assumption aligns with recent work on team contracts with peer effects, including [Claveria-Mayol et al. \(2024\)](#) and [Dasaratha et al. \(2025\)](#).

teams according to a common endogenous ordering. Rank projects by decreasing marginal value of quality to the principal in the optimum. Each worker's set of assigned and fully utilized projects is sufficiently characterized by two worker-specific project indices, ϕ_i and α_i , on that ordering. He is fully utilized at the first ϕ_i projects, assigned but underutilized at every project between ϕ_i and α_i , and left off projects beyond α_i (Theorem 1). While the ordering is endogenous in general, I identify conditions under which it can be read off the model primitives: the principal's valuation must rank projects consistently and workers must not differ in which projects they prefer (Proposition 5).

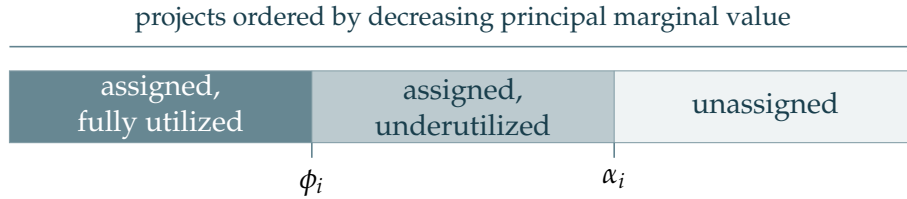


Figure 1: Optimal Worker Project Assignments

That optimal project assignments take an ordered structure is surprising given the heterogeneity in worker attributes. One expects the principal to exploit workers' comparative advantages and concentrate them on projects they prefer, forming distinct teams with strong complementarity. Assignments may overlap irregularly, if at all. Theorem 1 rules this out.

The intuition comes from an arbitrage. Suppose a worker is fully utilized at a low-value project but underutilized at a high-value one. Moving a small amount of his effort from the former to the latter is profitable for the principal, but it is infeasible. Reducing his effort on the low-value project causes a loss in complementarity for the teammates he leaves behind, jeopardizing their willingness to work there.

The key insight is to coordinate the profitable transfer across the *entire project team*. The principal cuts each teammate's cap at the low-value project by exactly the complementarity he lost and redirects his extra capacity toward the high-value project. This adjustment requires additional rounds: each cut reduces complementarity for that worker's own collaborators, whose caps must be adjusted in turn, and so on throughout the network. The same condition delivering a unique

equilibrium, which bounds complementarity relative to the curvature of effort costs, ensures that the series of adjustments converges. The limiting transfer is well-defined and read off a matrix capturing worker interactions within a project team. Qualitatively, workers with stronger ties to the underutilized worker exhibit a more pronounced shift in effort than those with weaker ties. Every worker's incentive constraints remain satisfied, so the transfer is both profitable and feasible. The optimum forbids such an arbitrage, so no worker is ever underutilized at a project the principal values more than one where he is fully utilized.

Which workers specialize in top-ranked projects, and which operate as generalists? While [Theorem 1](#) orders each worker's own projects, it does not order workers relative to each other, and that ordering is endogenous. Whether a worker should be assigned or fully utilized at a project depends on (1) who else is staffed there, which is determined by every other worker's cutoffs and (2) how thinly he is already spread, which is dictated by his own cutoff. Because costs and complementarity act through both channels at once, I separate them by shutting down each source of endogeneity in turn.

I first assume that complementarities between all workers are homogeneous so that the identity of a collaborator does not matter for a worker's marginal benefit. Costs and preferences remain heterogeneous, and what differentiates workers is their relative cost of concentrating effort on one project versus spreading it across multiple projects. A worker's *specialization index*, a statistic in the primitives, is increasing in his multitasking cost and decreasing in his cost of concentration and orders workers completely. A worker with a higher specialization index is assigned to and fully utilized on a subset of the projects to which any worker with a lower index is assigned ([Theorem 2](#)). That is, specialist project assignments are nested within those of generalists.

Next, I restore heterogeneous complementarity and assume the principal assigns every worker to every project. Every project team is identical, and I focus on the full utilization margin. A natural conjecture is that the most central workers should be fully utilized at the most projects. Because they collaborate well, letting them contribute freely everywhere amplifies efforts throughout the organization. However, this backfires. Fully utilizing a central worker at many projects spreads him thin and weakens his contribution to each team. The links that make him cen-

tral to the organization are the same links that make it expensive to fully utilize him across multiple projects.

What sorts workers is the *scope index*, the ratio of two distinct Katz-Bonacich centralities computed on the same network of complementarities but with different cost weightings ([Theorem 3](#)).⁷ The first measures a worker's influence on other workers' total effort. It is high when multitasking is cheap for him and for his close collaborators. The second measures his influence on other workers' effort within a project, and it is high when multitasking is expensive for him and for his close collaborators. Since both centrality measures aggregate the same collaboration links, a well-connected worker could score high on both and neither discerns whether a worker is better suited to be a specialist or generalist. Only the ratio sorts and workers with higher scope indices are fully utilized at a superset of the projects of workers with lower scope indices.

Finally, I fix an optimal project assignment and derive its comparative statics. Because incentive constraints are linear in effort, each one can be interpreted as a budget: a worker's preference for a project is the income of effort available to the principal and his complementarity and cost parameters are prices. The principal faces one budget per binding incentive constraint rather than a single budget. I show that shocks admit a Slutsky decomposition ([Theorem 4](#)). The substitution effect tracks movements "along" effort budgets and is only present when the principal underutilizes workers. When every assigned worker is fully utilized, any reallocation violates an incentive constraint that was previously binding, and the substitution effect vanishes ([Proposition 7](#)).

The network plays a starring role in the income effect, and I separate the latter into two opposing channels ([Proposition 8](#)). Consider an increase in a worker's preference for a project. He works more there, generating additional complementarity that induces his teammates to invest more in the project. This manifests in a *within-project spillover*. However, the extra effort raises their marginal costs of effort, so each worker reduces effort at other projects, leading to a loss in complementarity. I call this second channel the *network reallocation effect*. If every worker is assigned to every project, the second channel degrades the qualities of all projects

⁷Without the assumption that every worker is assigned to every project, the sorting index has no closed form. It retains the same ratio form but is built from endogenous multipliers and recomputed on a shrinking network as workers are dropped from projects ([Appendix D](#)).

not subject to the initial shock. In general, its effect on other projects depends on the pattern of assignment and the route of shock transmission ([Proposition 9](#)).

I see three main contributions. First, I offer an optimality rationale for ordered worker project assignments. They follow the same project ranking, and so project teams are nested on both the assignment and full utilization margins. Second, I show that a standard measure of centrality does not sort workers into generalists and specialists. While Katz-Bonacich centrality ranks workers by the effort they encourage, the relevant question is how much of that effort survives after a central worker is spread thin. The correct object is a ratio of two centralities capturing that trade-off. Third, I derive comparative statics for a network that the principal designs. The response to a shock admits a Slutsky decomposition, and the income effect decomposes into two opposing network effects with clear interpretations. A positive shock at one project can degrade the quality of every other project in the organization.

Related Literature

This paper contributes to three areas of research: network games, principal-agent theory and organization design.

Network games. Early theories of network games restrict workers to choosing a single action with complementary or substitutable interactions between neighbors ([Ballester et al., 2006](#); [Bramoullé and Kranton, 2007](#); [Bramoullé et al., 2014](#)). Two recent papers, [Claveria-Mayol et al. \(2024\)](#) and [Dasaratha et al. \(2025\)](#), embed a network game within a moral-hazard problem. The principal in both papers faces a single project, provides incentives through pay and takes the set of active agents as given. In contrast, I analyze how a principal sets effort caps at multiple projects and explicitly characterize where workers are active.

Work on network interventions posits a social planner who removes players or designs a vector of marginal benefits to influence activity ([Belhaj and Deroïan \(2018\)](#); [Galeotti et al. \(2020, 2025\)](#)). By determining how much complementarity workers transmit within a project, the principal in my model effectively selects which links within a network are active, and to what extent. The network is designed, whereas in the network formation literature, link formation is decentral-

ized (e.g. [Jackson and Wolinsky \(1996\)](#); [Bala and Goyal \(2000\)](#)).

In multiplex network games, players choose multiple actions across several layers subject to an aggregate effort constraint ([Chen et al. \(2018\)](#); [Zenou and Zhou \(2026\)](#)). Closest to this work is [Sadler and Yeh \(2026\)](#), in which players select neighbor-specific efforts subject to a convex cost of total effort; my effort game is a parametric multigraph version of it. The novelty here is that the principal selects the structure of the effort game, so the comparative statics pertain to those of an endogenous network. In a separate project, I study modular task assignments where effort at a task is a strategic *substitute* for effort at locally related tasks ([Yeh, 2026](#)). Here, efforts within a project are strategic complements for each other.

Principal-agent theory. Principal-agent models have focused on incentive design in multitask or multi-agent settings ([Itoh, 1991](#); [Holmstrom and Milgrom, 1991](#); [Devatripont et al., 2000](#); [Che and Yoo, 2001](#)). This paper combines both dimensions but replaces the contracting instrument with assignment.⁸

Assignment is closest to delegation. Setting an effort cap is akin to choosing a delegation set ([Alonso and Matouschek, 2008](#)), as applied in project choice ([Armstrong and Vickers, 2010](#)) and multidimensional actions ([Frankel, 2014](#)). Delegation is driven by asymmetric information, and the principal balances granting an agent flexibility to use his private information while limiting undesirable actions resulting from agent bias. My setting has no private information, and I identify a distinct reason for restricting action sets. Because effort costs are convex, forbidding it at one project allows the principal to obtain more effort at another. The delegation sets in my environment are also interdependent. Lowering a worker’s effort cap at one project shifts his action set at every other project through multitasking costs and his teammates’ action sets at that project through complementarity.

Organization design. In assignment and matching models with transferable utility, agents are typically sorted by a single scalar attribute ([Becker, 1973](#); [Kremer, 1993](#)). With a collaboration network and multitasking costs, what sorts workers in my model is a ratio of two network centralities. Closest is [Freund \(2025\)](#), who show that specialization within a firm generates coworker complementarity and

⁸Assignment problems are prominent in operations research. Allocation is chosen subject to capacity constraints for, say, a server, rather than to workers’ willingness to contribute effort. See [Gurvich and Van Mieghem \(2015, 2018\)](#) for examples and [Silva et al. \(2021\)](#) for a recent survey.

concentration of talent within a small number of firms. Workers belong to a single firm, and so the question of multi-team membership is moot.⁹ My main structural result also connects to optimal hierarchies, which arise from limited managerial attention (Geanakoplos and Milgrom, 1991), limited decision-making capacity (Hart and Moore, 2005), communication (Visser, 2000; Ambrus et al., 2013) and grouping activities for synergy (Dessein et al., 2010). These are hierarchies of authority or information. In contrast, the nested project assignments in the optimum are a hierarchy of team membership.

2 An Illustrative Example

I showcase the economics of assignments through an example with three workers $\{1, 2, 3\}$ and three projects $\{q, s, t\}$. I build the model in three stages to isolate each force: preferences first, then multitasking costs, then collaboration.

Worker i 's payoff from project p is linear in his effort e_i^p and quadratic in cost.

$$U_i^p = \beta_i^p e_i^p - \frac{c_i}{2} (e_i^p)^2$$

	Preference $\beta_i^{(\cdot)}$			Cost
	q	s	t	c_i
Worker 1	0.96	0.46	0.91	0.70
Worker 2	0.99	0.50	0.87	0.67
Worker 3	0.65	0.57	0.95	0.79

Table 1: Model Primitives

His intrinsic preference for working on project p is $\beta_i^p > 0$ and his cost of effort is scaled by $c_i > 0$. His total payoff is $U_i := \sum_p U_i^p$. The quality of a project is the total effort it receives, denoted by $Q^p := \sum_i e_i^p$, and the principal values quality according to a log-linear Cobb-Douglas function

$$V = 0.87 \ln(Q^q) + 0.99 \ln(Q^s) + 0.14 \ln(Q^t)$$

The principal's instrument is an effort cap. She sets a ceiling $\bar{e}_i^p \geq 0$ for each

⁹For additional work on the optimal composition of a team and how it is affected by size, diversity, and incentives, see Prat (2002); Georgiadis (2015); Fu et al. (2016); Glover and Kim (2021).

worker-project pair (i, p) and the worker then chooses effort across all projects to maximize his total utility subject to $e_i^p \leq \bar{e}_i^p$. The principal can hold a worker back, but she cannot compel him to work. At this stage, each worker's problem is additively separable across projects, so a cap at one project does not affect his effort provision anywhere else. Even though the principal and her workers prefer different projects, it is optimal to set a cap at or above the effort level that solves a worker's first-order condition i.e. $\bar{e}_i^p \geq \beta_i^p / c_i$. The assignment problem is trivial; multitasking costs and collaboration are what make it interesting.

Multitasking. Workers incur multitasking costs from working on multiple projects.

$$U_i^p = \beta_i^p e_i^p - \frac{c_i}{2} (e_i^p)^2 - \frac{r_i}{2} \sum_{q \neq p} e_i^p e_i^q$$

	r_i
Worker 1	0.13
Worker 2	0.17
Worker 3	0.25

Table 2: Multitasking Costs

Effort at one project raises the marginal cost of effort at every other project by $r_i > 0$. Multitasking costs make the structural misalignment in the preferences between the principal and workers a genuine problem. Panel A of [Table 4](#) shows what workers do if they are left unconstrained i.e. $\bar{e}_i^p = \infty$. Each worker invests heavily in the project he cares most about (q or t), and exerts minimal effort at project s . This is suboptimal for the principal, who likes s the most.

Panel B of [Table 4](#) gives the optimum. The principal uses caps to direct effort toward project s . She removes workers 2 and 3 from project t , which frees up capacity for them to work at project s ; their efforts nearly double there. Worker 1 is capped below what he would freely contribute, as his marginal benefit there strictly exceeds his marginal cost. Even though worker 3 prefers project t the most, he is taken off of it because his multitasking cost is much greater than those of others. The principal would rather he work on more valuable projects like q and s .

Collaboration. Now, workers enjoy working with teammates.

The complementarity terms $g_{ij} = g_{ji} > 0$ measure how well workers i and j work together. Workers 1 and 3 are the strong pair. Panel C of [Table 4](#) shows how

$$U_i^p = \beta_i^p e_i^p + \sum_{j \neq i} g_{ij} e_i^p e_j^p - \frac{c_i}{2} (e_i^p)^2 - \frac{r_i}{2} \sum_{q \neq p} e_i^p e_i^q$$

	g_{ij}
Workers (1, 2)	0.04
Workers (1, 3)	0.13
Workers (2, 3)	0.03

Table 3: Complementarities

collaboration changes the optimal staffing pattern. What matters is no longer just how well a worker multitasks, but who he multitasks alongside. Workers 1 and 3 are the central pair, and one might expect the principal to place them at all three projects to exploit their outsized complementarities. In the optimum, she does the opposite: they are assigned to projects q and s and removed from t . Worker 2, who has the second-highest multitasking cost but collaborates weakly with both of them, covers t alone and ends up being the generalist.

	Multitasking Costs Only						With Collaboration		
	A. Unconstrained			B. Optimum			C. Optimum		
	q	s	t	q	s	t	q	s	t
Worker 1	1.120	0.242	1.041	1.151	0.273	0.842[†]	1.475	0.513	0
Worker 2	1.227	0.298	0.993	1.393	0.463	0	1.372	0.361	0.849
Worker 3	0.426	0.278	0.990	0.664	0.516	0	0.649[†]	0.623	0
Quality $Q^{(\cdot)}$	2.773	0.818	3.024	3.208	1.252	0.842	3.496	1.497	0.849

[†] Marginal benefit strictly exceeds marginal cost.

Table 4: Worker Efforts and Project Quality

This example leaves two questions. Do optimal assignments take on a particular structure, and which workers does the principal designate as specialists, and which as generalists? The main analysis answers both.

3 Model

There is one principal (she), a set of workers (he) $N = \{1, \dots, n\}$ and a set of projects $P = \{1, \dots, m\}$.

Workers. Worker i chooses nonnegative effort $e_i^p \in [0, \bar{e}_i^p]$ to exert at each project p where $\bar{e}_i^p > 0$. He receives utility given by the sum of per-project benefits less his total cost of effort.

$$U_i(\mathbf{e}_i, \mathbf{e}_{-i}) = \underbrace{\sum_{p=1}^m (\beta_i^p e_i^p + \sum_{j \neq i} g_{ij} e_i^p e_j^p)}_{\text{per-project benefit}} - \underbrace{\left(\sum_{p=1}^m \frac{c_i}{2} (e_i^p)^2 + r_i \sum_{p,q \mid p < q} e_i^p e_i^q \right)}_{\text{cost of effort}}$$

where $\mathbf{e}_i = \{e_i^p\}_{p=1}^m$ and $\mathbf{e}_{-i}$ represents other workers' efforts at all projects.

From project p , worker i receives a benefit that is linear in his effort. His intrinsic preference for working on project p is captured by $\beta_i^p > 0$. Effort complementarity between worker i and j is measured by $g_{ij} > 0$, which I assume to be symmetric i.e. $g_{ij} = g_{ji}$.

Worker i incurs a general quadratic cost of effort. His cost of specialization on a single project is $c_i > 0$, while his cost of multitasking is $r_i > 0$. To maintain convexity in costs, I assume that $c_i > r_i$.¹⁰ Because the cost of effort is convex and benefit is linear, a worker's utility is concave in his own effort vector.

Principal. The quality of project p is given by the sum of worker efforts $Q^p := \sum_{i=1}^n e_i^p$. The principal values project qualities according to $V : \mathbb{R}^m \rightarrow \mathbb{R}$, which is strictly increasing in each argument, strictly concave and twice continuously differentiable. The principal's choice of upper bounds $\bar{\mathbf{e}} \in \mathbb{R}_+^{mn}$ solves:

$$\max_{\bar{\mathbf{e}} \geq 0} V(Q^1, \dots, Q^m)$$

s.t. $\mathbf{e}$ is the (principal-preferred) pure-strategy Nash equilibrium given $\bar{\mathbf{e}}$

If there are multiple equilibria induced by $\bar{\mathbf{e}}$, then the one that achieves the highest principal valuation is selected. Subsequently, I provide conditions under which there is always a unique equilibrium so no selection rule is required.

¹⁰The Hessian of worker i 's cost of effort is $(c_i - r_i)\mathbf{I}_m + r_i\mathbf{1}_m\mathbf{1}_m^\top$, which has eigenvalues $c_i + (m - 1)r_i$ and $c_i - r_i$ with multiplicity of $m - 1$. For the cost to be strictly convex, its Hessian needs to be positive definite, which is true if $c_i - r_i > 0$.

3.1 Implementability

To characterize effort profiles that are implementable as equilibria, I exploit symmetric effort complementarity to cast the workers' effort game as an exact potential game. Nash equilibria of the effort game correspond to maxima of the exact potential (Monderer and Shapley, 1996), which is given by

$$\Psi(\mathbf{e}) := \sum_{i=1}^n U_i(\mathbf{e}) - \frac{1}{2} \sum_{i=1}^n \sum_{j \neq i} \sum_{p=1}^m g_{ij} e_i^p e_j^p.$$

If its Hessian matrix is negative definite, then the exact potential is strictly concave and has a unique maximizer corresponding to a unique Nash equilibrium. The negative Hessian, which I call the *worker interaction matrix* $\mathbf{W}$, captures how worker interactions affect the marginal cost of effort.¹¹ To illustrate, suppose there are two workers i and j and two projects p and q . The worker interaction matrix is

$$\mathbf{W} = \begin{matrix} & \begin{matrix} (i,p) & (i,q) & (j,p) & (j,q) \end{matrix} \\ \begin{matrix} (i,p) \\ (i,q) \\ (j,p) \\ (j,q) \end{matrix} & \begin{bmatrix} c_i & r_i & -g_{ij} & 0 \\ r_i & c_i & 0 & -g_{ij} \\ -g_{ji} & 0 & c_j & r_j \\ 0 & -g_{ji} & r_j & c_j \end{bmatrix} \end{matrix}.$$

Working on project p raises worker i 's marginal cost at project p by his specialization cost c_i , listed in $\mathbf{W}[(i,p), (i,p)] = c_i$. It also increases his marginal cost at project q by $\mathbf{W}[(i,p), (i,q)] = r_i$. An increase in worker j 's effort on project p raises worker i 's marginal benefit, or equivalently reduces his marginal cost there, represented by $\mathbf{W}[(i,p), (j,p)] = -g_{ij}$. There are no explicit interactions between different workers at distinct projects i.e. $\mathbf{W}[(i,p), (j,q)] = 0$ because synergies between workers operate within a project, and multitasking costs affect a single worker across multiple projects. Nevertheless, these interactions surface in equilibrium. Block matrix multiplication allows us to write $\mathbf{W}$ in terms of three $n \times n$ matrices capturing specialization, multitasking, and complementarity.

Definition 1. The **worker interaction matrix** is $\mathbf{W} := (\mathbf{C} \otimes \mathbf{I}_m + \mathbf{R} \otimes (\mathbf{1}_m \mathbf{1}_m^\top - \mathbf{I}_m)) - (\mathbf{G} \otimes \mathbf{I}_m)$ where the **specialization cost matrix** is $\mathbf{C} := \text{diag}(c_i)_{i \in N}$, the **multitask-**

¹¹Defining it as the *negative* Hessian is convenient for writing equilibrium efforts later.

ing cost matrix is $\mathbf{R} := \text{diag}(r_i)_{i \in N}$, the **complementarity matrix** is $\mathbf{G}$ with entries $G_{ij} = g_{ij}$ and 0 on its diagonal, and $\otimes$ is the Kronecker product.¹²

Assumption 1. *The worker interaction matrix $\mathbf{W}$ is positive definite.*

The Hessian of the exact potential is negative definite, so for a given vector of effort caps, there is a unique Nash equilibrium in the effort game, and no selection rule is needed in the principal's problem. To simplify the principal's problem, I show that it is without loss of generality for the principal to directly choose efforts so long as each worker's *incentive constraint* holds whenever effort is positive.

Proposition 1. *For $\mathbf{e} \geq \mathbf{0}$, there exists an upper bound $\bar{\mathbf{e}}$ that implements $\mathbf{e}$ as an equilibrium if and only if*

$$\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q$$

whenever $e_i^p > 0$. Moreover, the principal can implement $\mathbf{e}$ by setting $\bar{\mathbf{e}} = \mathbf{e}$.

Proof. See Appendix A.1. □

When a worker's marginal utility from working on a project is nonnegative, reducing effort is never profitable. Capping effort at the principal's target level prevents the worker from contributing more even if he would like to.

The listed inequality is worker i 's **incentive constraint** at project p . The left term is worker i 's marginal benefit at project p . It depends on his intrinsic preference and the efforts of other workers at that project. An increase in other workers' efforts at project p relaxes his incentive constraint. The right term is worker i 's marginal cost at project p , which is determined by the efforts he supplies at other projects. Raising his effort cap at project p will not encourage more effort if his incentive constraint binds. Instead, the principal has to lower his effort caps on other projects or increase the efforts of his collaborators on project p .

¹²Formally, given a $m \times n$ matrix $\mathbf{A}$ and a $p \times q$ matrix $\mathbf{B}$, the Kronecker product $\mathbf{A} \otimes \mathbf{B}$ is a $mp \times nq$ block matrix given by $\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{m1}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{bmatrix}$.

3.2 Set-up: A Convex Optimization Problem

Implementing zero effort at a project requires no incentive constraint. It is important to disentangle the reasons why a worker exerts zero effort in equilibrium: (1) despite his willingness to contribute, he is constrained by an effort cap of zero or (2) he is not willing to contribute there due to a negative marginal utility at zero effort. In the second scenario, the worker is strictly worse off when starting work on a project, which falls outside the scope of this paper. To reiterate, I analyze when the principal restricts activity that both she and a worker value. Ruling out the second scenario also keeps the analysis tractable; otherwise, we need to track whether an incentive constraint needs to be satisfied or can be safely ignored at zero effort. The principal's problem becomes combinatorially difficult due to the potential nonconvexity of the feasible set.¹³

Accordingly, I restrict parameters so that workers are always motivated to start work on a project, and are forced off a project only if the principal sets their effort cap to zero. That way, the set of implementable effort profiles coincides with the set of profiles satisfying worker-project incentive constraint, including those at zero effort. What must hold is worker i 's incentive constraint at zero effort on project p :

$$\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq r_i \sum_{q \neq p} e_i^q.$$

To satisfy it, his multitasking costs r_i cannot be too high relative to his intrinsic preference and his total effort. The latter is endogenous, and I bound it by summing his incentive constraints across projects and collecting terms in total effort, which I denote as $E_i := \sum_{p=1}^m e_i^p$

$$\sum_{p=1}^m \beta_i^p + \sum_{j \neq i} g_{ij} E_j \geq (c_i + (m-1)r_i) E_i$$

Regardless of where a worker allocates it, raising his total effort incurs a unit increase in specialization cost and a $(m-1)$ -unit increase in multitasking costs. A rise in worker j 's total effort reduces worker i 's marginal cost by g_{ij} . Stacking across

¹³Appendix B formulates the principal's problem as a mixed-integer nonlinear program and gives an explicit example of a nonconvex feasible set.

all workers gives a system in total efforts governed by the *aggregate interaction matrix*.

Definition 2. The **aggregate interaction matrix** is $\mathbf{P} := \mathbf{C} + (m - 1)\mathbf{R} - \mathbf{G}$.

The entries of $\mathbf{P}$ map changes in *total* effort to changes in a worker's marginal cost of *total* effort. The stacked inequalities read $\mathbf{P}\mathbf{E} \leq \sum_{p=1}^m \beta^p$, so any effort profile satisfying every incentive constraint obeys $\mathbf{E} \leq \tilde{\mathbf{E}} := \mathbf{P}^{-1} \sum_{p=1}^m \beta^p$.¹⁴

Assumption 2. For each worker i , $r_i \tilde{E}_i < \min_p \beta_i^p$.

A worker's total multitasking burden never outweighs his preference for his least favorite project, guaranteeing he is always strictly motivated to start work at any project. When preferences are uniform i.e. $\beta_i^p \equiv \beta$ for all (i, p) , [Assumption 2](#) reduces to a simpler condition in the primitives: $c_i - r_i \geq r_i(\sum_{j \neq i} g_{ij}/r_j)$. While the model already assumes $c_i - r_i > 0$, this condition asks for more. The cost of specialization must be high and that of multitasking low enough to dominate the pull from collaborators who are good at multitasking. The assumption rules out parameters under which a worker would rather work a lot on one project alongside his strongest collaborators than start on a new one. Most importantly, the assumption does not constrain how workers differ, such as who multitasks well, who collaborates better with whom, and which projects workers like most or least. With this, the principal solves a tractable convex optimization problem.

Proposition 2. Under [Assumption 2](#), every implementable effort profile satisfies $\mathbf{E} \leq \tilde{\mathbf{E}}$, implying the set of implementable effort profiles coincides with the set of effort profiles that satisfy incentive constraints for all worker-project pairs. The principal's problem is:

$$\begin{aligned} & \max_{\mathbf{e}} V(Q^1, \dots, Q^m) \\ & \text{s.t. } \mathbf{e} \geq \mathbf{0}, \\ & \forall (i, p), \beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q \\ & \forall p, \sum_{i=1}^n e_i^p = Q^p \end{aligned}$$

Proof. See [Appendix A.2](#). □

¹⁴I show in [Appendix A.2](#) that $\mathbf{P}^{-1}$ exists and is nonnegative.

A solution to the principal's problem exists because the objective is continuous and the feasible set is closed and bounded. Moreover, the solution is unique in total efforts because the objective is strictly concave over total efforts and the feasible set is convex.

3.3 Remarks

1. There are no explicit contracts in this model. In [Section 7.1](#), I consider an alternate model where the principal rewards effort. To implement the same outcomes as effort caps, the principal needs to use *negative* wages.
2. Symmetric complementarity is sufficient for a unique equilibrium. Asymmetric complementarities, discussed in [Section 7.2](#), do not affect the main results.
3. In the model, there is no physical upper limit on workers' efforts. The main results remain robust to imposing an external hard limit on efforts (for example, a 40-hour work week with $\sum_p e_i^p \leq T_i$) as shown in [Section 7.3](#).

4 Structure of Optimal Project Assignments

To describe staffing patterns in optimal project assignments, I classify effort caps based on two properties.

Definition 3. In effort profile $\mathbf{e}$, worker i is **assigned** to project p if $e_i^p > 0$, otherwise he is **unassigned**. Worker i is **fully utilized** at project p if his incentive constraint binds i.e.

$$\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p = c_i e_i^p + r_i \sum_{q \neq p} e_i^q.$$

He is **underutilized** if

$$\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p > c_i e_i^p + r_i \sum_{q \neq p} e_i^q.$$

The **assignment** and **full utilization sets** for project p are $A^p := \{i \mid e_i^p > 0\}$ and $\Phi^p := \{i \mid i \text{ is fully utilized at } p\}$.

Underutilization prevents a worker from contributing as much as he wishes. While unassigning worker i from project p eliminates p as a source of multitasking costs for other projects, it forfeits the complementarity he generates for other workers at project p . By assigning but underutilizing him, the principal leverages him as a source of complementarity for others on a particular project while preserving his capacity to work on other projects.

4.1 Ordered Staffing Patterns

Is there a pattern to which workers are assigned and fully utilized? The main structural result demonstrates that both assignment and utilization are ordered by a common endogenous ranking. Before describing the result, I list two required generic conditions of the optimum.

Assumption 3. *In the optimum, (1) every binding incentive and nonnegativity constraint has strictly positive multipliers and (2) worker i 's incentive constraint at project p is slack whenever $e_i^p = 0$.*

Condition (1) assumes strict complementarity at the optimum. For condition (2) to fail, there must be a worker-project pair (i, p) where the incentive constraint exactly binds at zero effort, which is a knife-edge case. Appendix C formally shows that both are generic. For convenience, denote $V^p := \partial V / \partial Q^p$.

Theorem 1. *Rank projects by the principal's marginal value in the optimum so that $V^1 \geq \dots \geq V^m$. For each worker i , there exist natural numbers $\phi_i, \alpha_i \in \{1, \dots, m\}$ where $\phi_i \leq \alpha_i$ such that he is*

- *fully utilized at projects $1, \dots, \phi_i$,*
- *assigned but underutilized at projects $\phi_i + 1, \dots, \alpha_i$,*
- *and not assigned to projects $\alpha_i + 1, \dots, m$.*

This implies the assignment and full utilization sets satisfy

$$\Phi^1 \supseteq \dots \supseteq \Phi^m \quad \text{and} \quad A^1 \supseteq \dots \supseteq A^m.$$

Proof. See Appendix [A.3](#). □

For each worker, two cutoffs determine which projects he is assigned to and fully utilized at. His project assignments comprise three successive regimes. For the most valuable projects, the principal allows him to work as much as he is willing to. For medium-value projects, the principal assigns him but underutilizes him. The worker would like to work more, but is prohibited from doing so. On the remaining low-value projects, he remains unassigned. All worker assignments agree on the *same* ranking of projects, namely by the principal's marginal value in the optimum. Project teams are necessarily nested. I give a diagnostic in the primitives for when unassigning or underutilizing workers is necessary in [Section 4.4](#). While an endogenous ordering is a consequence of the model's generality, I provide conditions in [Section 4.5](#) such that project ordering is exogenously determined.

That optimal project assignments embody such a clean structure is surprising, especially because all worker attributes— preferences, synergies, and costs— are heterogeneous. One may expect workers to retain a comparative advantage at certain projects over others, and it seems natural that workers specialize in distinct projects and have non-overlapping project assignments. [Theorem 1](#) precludes any worker from being underutilized or excluded from a project while being fully utilized or assigned to a less valuable one.

Why then should worker assignments be ordered this way? There are two pieces to the puzzle. Between projects the principal values differently on the margin, the ordering follows from a transfer of efforts that would otherwise be profitable ([Section 4.2](#)). Suppose a worker is assigned to a low-value project but would like to work more at a high-value one. Moving his efforts alone is infeasible because it reduces complementarity at the project team he leaves behind. Instead, I design a *team-coordinated* effort transfer from the low-value project to the high-value one that is both feasible and profitable. For projects with the same marginal valuation, I show that they have the same assignment and full utilization sets ([Section 4.3](#)). These “plateaus” correspond to the projects where workers are assigned but underutilized.

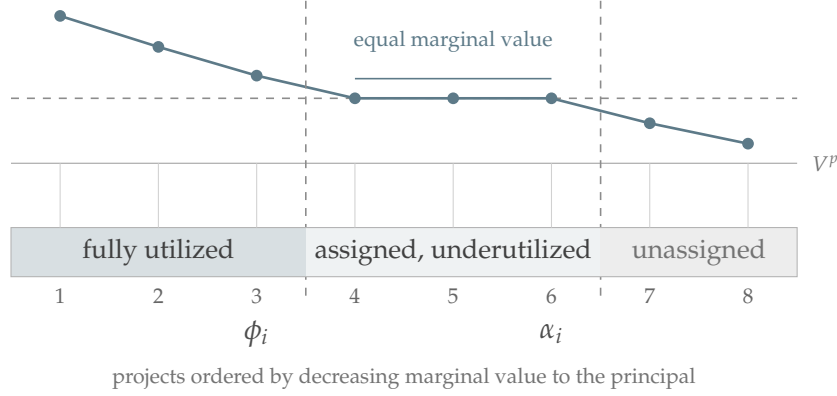


Figure 2: Optimal Worker Project Assignment

4.2 A Team-Coordinated Effort Transfer

To ensure that an effort transfer for workers between projects with different marginal values is feasible, I track the **slack** (s_i^p) of an incentive constraint, which must always remain nonnegative. Substituting total effort $\sum_{q \neq p} e_i^q = E_i - e_i^p$ separates how effort at project p and total effort affect slack there.

$$s_i^p = \beta_i^p + \sum_{j \neq i} g_{ij} e_j^p - (c_i - r_i) e_i^p - r_i E_i.$$

Suppose there are two projects p and q with $V^p > V^q$ in the optimum and worker i is fully utilized at q , but underutilized at p , so he has positive slack at the latter. Consider the naïve transfer that moves $\delta > 0$ of his effort from q to p , which is profitable for small enough δ because

$$\Delta V = (V^p - V^q)\delta + O(\delta^2).$$

Is it feasible? This transfer keeps worker i 's total effort constant, so it leaves his slack at every other project unchanged. At project p , his slack falls by $(c_i - r_i)\delta$, which is absorbed by his existing slack. Every other worker's slack at p increases from the extra complementarity. The problem arises at project q . Reducing worker i 's effort decreases complementarity at q for his teammates there, cutting their slack and potentially violating their incentive constraints.

The key insight is to coordinate the effort transfer on the entire project team

at q rather than just worker i . For simplicity, I assume for now that every worker is assigned to project q i.e. $A^q = N$; I return to the case when $A^q \subset N$ later. I design a transfer $\mathbf{x} \in \mathbb{R}^n$ that moves worker j 's effort from q to p by x_j satisfying two requirements: (1) worker i 's slack at p decreases by $\delta > 0$ and (2) every other worker's slack at both projects remains unchanged. Because the transfer preserves total effort, the change in slack is given by

$$\Delta s_j^p = \sum_{k \neq j} g_{jk} \Delta e_k^p - (c_j - r_j) \Delta e_j^p = -\Delta s_j^q.$$

The two requirements put together give $\mathbf{M}\mathbf{x} = \delta \mathbf{e}_i$ where $\mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$ and $\mathbf{e}_i$ is the i -th basis vector.¹⁵ Rather than invert $\mathbf{M}$ directly, I build intuition for what this transfer scheme achieves by leveraging a "round-by-round" accounting of how each worker's slacks change every time effort for another worker is transferred.

Round 1: The principal only transfers worker i 's effort by $x_i^{(1)} = \delta / (c_i - r_i)$. This delivers the $-\delta$ change in his slack at p . Notice that for every other worker $j \neq i$, his slack at p expands by $g_{ji}x_i^{(1)}$ while his slack at q contracts by the same amount.

Round 2: The principal transfers efforts for every other worker $j \neq i$ to absorb his extra slack at p and erase his deficit at q , given by

$$x_j^{(2)} = \frac{g_{ji}}{c_j - r_j} x_i^{(1)}$$

This transfer creates additional slack at p and deficit at q for j 's collaborators, which calls for a third round of transfers, and so on. The vector of transfers for each round satisfies a recurrence relation $\mathbf{x}^{(t)} = (\mathbf{C} - \mathbf{R})^{-1} \mathbf{G} \mathbf{x}^{(t-1)}$. That is, effort in round t travels along walks of length $t - 1$ within the network of complementarities, weighted by complementarity g_{ji} and discounted by each worker's net cost $c_j - r_j$ of shifting effort between projects holding total efforts fixed. Summing over all adjustments (or walks) gives

$$\mathbf{x} = \sum_{t=1}^{\infty} ((\mathbf{C} - \mathbf{R})^{-1} \mathbf{G})^{t-1} (\mathbf{C} - \mathbf{R})^{-1} \delta \mathbf{e}_i$$

[Assumption 1](#) prevents complementarity from causing the adjustment process to

¹⁵Recall $\mathbf{C} := \text{diag}(c_i)_{i \in N}$, $\mathbf{R} := \text{diag}(r_i)_{i \in N}$, and $\mathbf{G}$ has entries $\mathbf{G}_{ij} = g_{ij}$ and 0 on its diagonal.

explode and rules out divergence. To state the formal result, I now give a name to $\mathbf{M}$, whose inverse is the limit of convergence.

Definition 4. The **team interaction matrix** is $\mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$.

The $\mathbf{M}$ matrix governs how worker interactions affect workers' marginal cost of effort at a project, *holding total efforts fixed*. The off-diagonals, which are $-g_{ij}$, capture complementarity between workers. The diagonal term $c_i - r_i$ measures the change in worker i 's marginal cost of effort at project p while keeping his total effort constant. An extra unit of effort at project p costs c_i , and to keep total effort constant, he must give up a unit of effort at another project q for a refund of r_i . This shift in effort *raises* his marginal cost at project p by $c_i - r_i$. The next result establishes that the series of effort adjustments converges to $\mathbf{M}^{-1}$.

Lemma 1. Under [Assumption 1](#), $\mathbf{M}$ has a nonnegative inverse that satisfies

$$\mathbf{M}^{-1} = \sum_{t=1}^{\infty} ((\mathbf{C} - \mathbf{R})^{-1} \mathbf{G})^{t-1} (\mathbf{C} - \mathbf{R})^{-1}.$$

Proof. See [Appendix A.4](#). □

The effort transfer has a simple formula: $\mathbf{x} = \mathbf{M}^{-1} \delta \epsilon_i$. Qualitatively, the principal reallocates a greater amount of effort for workers with stronger ties to worker i . It is profitable for sufficiently small δ because

$$\Delta V = (V^p - V^q) \mathbf{1}_n^T \mathbf{x} + O(\|\mathbf{x}\|^2)$$

In the general case where $A^q \subset N$, the matrix of interest is $\mathbf{M}^{(A^q)}$, which is the submatrix of $\mathbf{M}$ with rows and columns restricted to only the workers on project team q . It inherits key properties of $\mathbf{M}$, including positive definiteness and a non-negative inverse, and so a similar argument succeeds. I leave the technical details to the main proof in [Appendix A.3](#).

Stated plainly, assigning a worker to a project with low marginal value but holding him back on one with high marginal value presents an arbitrage. The strategy is to redistribute the entire project team's effort towards the high-value project while respecting team members' incentive constraints. The mathematical

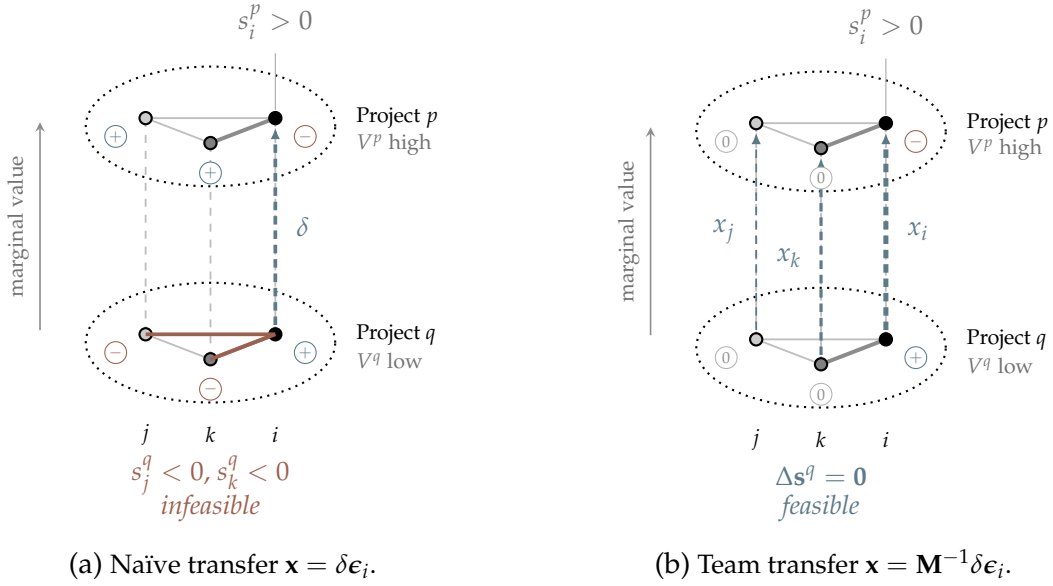


Figure 3: Worker i and team-coordinated transfers where symbols $(+)$, (0) and $(-)$ indicate whether slack increases, stays the same or decreases.

object of interest is the team interaction matrix $\mathbf{M}$, which pins down how the principal should design such an effort transfer.

4.3 Projects with Equivalent Marginal Values

Projects with the same marginal value in the optimum obey nested staffing patterns; in fact, these projects necessarily have identical assignment and full utilization sets.

Proposition 3. *Projects with equal marginal value in the optimum have identical assignment and full utilization sets. Moreover, each worker's underutilized projects share a common marginal value $V^{\phi_i+1} = \dots = V^{\alpha_i}$.*

Proof. See Appendix A.5. □

Two projects with the same marginal value pose the same staffing problem to the principal. Positive definiteness of the team interaction matrix ensures the marginal cost of effort accrued by a worker outweighs the collaboration bonus he receives. This guarantees a unique solution to the staffing problem by ruling out

coordination failures that lead to multiple assignment/full utilization solutions. The second statement can be directly shown using a transfer argument. Suppose the principal underutilizes a worker at two projects, p and q , in the optimum. She can perform the same team-coordinated effort transfer as before if there is an imbalance in marginal values. For example, if $V^p > V^q$, then she can pull efforts from q towards p for a profitable deviation. Therefore, it must be that both projects share the same marginal value.

It is tempting to read [Proposition 3](#) as making the interval of projects with underutilization fragile because they must have equivalent marginal values. This is inaccurate: the effort for an underutilized worker-project pair is defined by two *strict* inequalities on the primitives— a slack incentive constraint and a slack non-negativity constraint. The example in [Section 2](#) has a unique optimum and so the underutilization pattern there survives small perturbations in every parameter.

4.4 When to Underutilize and Unassign

While [Theorem 1](#) explains how assignment and utilization cutoffs are ordered, it does not describe when either staffing margin is required in the optimum. As a benchmark, suppose the principal sets no caps at all: $\bar{e}_i^p = \infty$ for all (i, p) . Workers exert effort in the unconstrained equilibrium $\hat{\mathbf{e}} := \mathbf{W}^{-1}\boldsymbol{\beta}$, which I call the **no-intervention outcome**. Because each row of the worker interaction matrix $\mathbf{W}$ corresponds to an incentive constraint, stacking them gives $\mathbf{W}\mathbf{e} \leq \boldsymbol{\beta}$. Every incentive constraint binds in $\hat{\mathbf{e}}$, which is guaranteed to be strictly positive by [Assumption 2](#).

Lemma 2. *Under [Assumption 2](#), $\hat{\mathbf{e}} > \mathbf{0}$.*

Proof. See [Appendix A.6](#). □

The no-intervention outcome is the unique assignment that assigns and fully utilizes every worker at every project. As we saw in the example of [Section 2](#), effort caps mitigate the misalignment between the principal’s valuation and the workers’ equilibrium contributions, with the latter shaped not only by workers’ intrinsic preferences, but also by the primitives governing collaboration and effort costs. I provide a diagnostic of when this misalignment occurs.

Proposition 4. *The no-intervention outcome is optimal if and only if $(\mathbf{W})^{-1}\nabla_{\mathbf{e}}V(\hat{\mathbf{e}}) \geq \mathbf{0}$.*

Proof. See Appendix A.7. □

The diagnostic is a computation in the primitives. If $(\mathbf{W})^{-1}\nabla_{\mathbf{e}}V(\hat{\mathbf{e}}) \geq \mathbf{0}$, then it is optimal to simply let the workers choose efforts as they wish. If the vector yields a negative entry, then the principal must unassign or underutilize in the optimum.

4.5 Exogenous Project Orderings and Cutoffs

The ordering of projects in Theorem 1 is endogenous, determined by how much effort a project inherently attracts weighted against its value to the principal. I provide a simple condition allowing us to predict how projects are ranked in the optimum.

Definition 5. The principal's valuation is **swap-increasing** if there exists an ordering of projects such that for all pairs of projects $p < q$ and every profile of project qualities $Q^p < Q^q$,

$$V(\dots, Q^p, \dots, Q^q, \dots) \leq V(\dots, Q^q, \dots, Q^p, \dots)$$

where only the p -th and q -th arguments are swapped.

Holding other project qualities fixed, the principal prefers higher quality at project p over project q whenever $p < q$. The log-linear Cobb-Douglas valuation in Section 2 is swap-increasing in the $\gamma^{(\cdot)}$ -weight ordering.

Swap-increasing valuations are insufficient to predict project orderings if workers' preferences are heterogeneous. Because some projects naturally attract high effort from workers, their marginal values could be pushed lower in the optimum. To prevent variations in worker preferences from causing unpredictable project reorderings, I assume uniform worker preferences across projects. Complementarity and cost parameters can remain heterogeneous. At the optimum, top-value projects under a swap-increasing valuation achieve maximum quality and team size.

Proposition 5. *If V is swap-increasing and $\beta_i^p \equiv \beta_i$ for all i , then $Q^1 \geq \dots \geq Q^m$ and $V^1 \geq \dots \geq V^m$ in the optimum.*

Proof. See Appendix A.8. □

Finally, obtaining a closed-form solution in the assignment and full utilization cutoffs requires the most restrictive parametric assumptions. In what follows, I assume the principal's valuation is log-linear Cobb-Douglas and workers are identical. Exploiting a structural symmetry in the workers allows us to pinpoint how the underutilization cutoff responds to changes in worker attributes.

Proposition 6. *Assume that all workers are identical and the principal's valuation is $V = \sum_{p=1}^m \gamma^{(p)} Q^p$ where $\gamma^{(1)} \geq \dots \geq \gamma^{(m)}$ and $\sum_{p=1}^m \gamma^{(p)} = 1$. There exists a worker-symmetric optimal project assignment with the following properties.*

1. *Every worker is assigned to every project.*
2. *There exists a single project index ϕ where every worker is fully utilized at projects $s \leq \phi$ and underutilized at projects $t > \phi$.*
3. *The ϕ index is given by*

$$\phi := \max \left\{ \hat{\phi} \text{ s.t. } \frac{\sum_{p=1}^{\hat{\phi}} \gamma^{(p)}}{\gamma^{(\hat{\phi})}} - \hat{\phi} \leq \frac{c - r - g(n-1)}{r} \right\}^{16}$$

4. *The project index ϕ is non-increasing in r , g and n , non-decreasing in c and independent of β .*

Proof. See Appendix A.9. □

Because the log-linear Cobb-Douglas satisfies Inada conditions at each project, every worker must be assigned to every project in the worker-symmetric optimum. The assignment margin shuts down, allowing us to isolate and focus on utilization. It is characterized by a single project index ϕ which has a closed-form solution.

¹⁶Under worker homogeneity, Assumption 1 is equivalent to $c > r + g(n-1)$ as $c - r - g(n-1)$ is one of the eigenvalues of $\mathbf{W}$.

All else equal, raising multi-tasking costs r or collaborative gains g makes it more valuable for the principal to concentrate more effort on a smaller set of projects. When workers struggle to work on many projects simultaneously, the principal limits their utilization on more projects, pushing them to exert higher effort on more valuable projects. When there are higher gains to collaboration, the principal extracts higher quality from more valuable projects at the expense of the less valuable ones. An increase in team size n likewise strengthens complementarities, and so it operates in the same way as an increase in g . A rise in c makes specialization more expensive, leading the principal to seek a more balanced project assignment. Finally, ϕ does not depend on β because β does not affect the cost of spreading or concentrating effort across projects.

5 Sorting Specialists and Generalists

Which workers specialize in top-ranked projects, and which operate as generalists? Which workers are fully utilized or underutilized at a project? In general, the ordering of the assignment and full utilization cutoffs is endogenous, and endogeneity arises from two different sources. The first is which collaborators a worker works with on a particular project. Whether to assign and/or fully utilize a worker at a project depends on which other workers are staffed there and so each worker's cutoff depends on every other worker's cutoffs. The second is how many projects a worker is on. Because a worker incurs multitasking costs, whether the principal should assign him to one more project depends on how thinly he is already spread. However, the latter is precisely what his assignment and full utilization cutoffs determine.

To understand how each force—complementarity and effort costs—sorts workers, I shut down each source of endogeneity in turn. In [Section 5.1](#), I assume complementarities are homogeneous, so that the identity of a collaborator is irrelevant to a worker's marginal benefit. I examine how heterogeneity in costs orders workers and derive a statistic in the primitives—the *specialization index*—that orders both workers' assignment and full utilization cutoffs. In [Section 5.2](#), I allow for heterogeneous complementarity but assume the principal assigns every worker to every project in the optimum. The *scope index*, which measures a worker's com-

parative advantage as a generalist over a specialist, orders workers' full utilization cutoffs. I leave the complete discussion of the general endogenous sorting index to Appendix D.

To be clear, I do not make any parametric assumptions on the principal's valuation function in this section. I also assume worker preferences are heterogeneous. As established in Section 4.5, while preferences influence the ordering of projects in the optimum, they do not affect the relative ordering of ϕ_i and α_i across workers. Variations in preferences do not alter the cost of concentrating or spreading a worker across multiple projects.

5.1 The Specialization Index

I assume workers differ in their specialization costs c_i , multitasking costs r_i and preferences β_i^p but have identical complementarity i.e. $g_{ij} \equiv g$ for all $i \neq j$. In this environment, the trade-off inherent to assigning and fully utilizing a worker at multiple projects is captured by the *specialization index*.

Definition 6. The **specialization index** of worker i is $\chi_i := r_i / (c_i + g)$.

The index reflects the relative cost of multitasking to specialization. A worker has a high specialization index if multitasking is expensive relative to specialization, and a low index in the reverse case. The following result shows that it sufficiently sorts generalists and specialists.

Theorem 2. *In the optimum, assignment and full utilization cutoffs α_i and ϕ_i are sorted such that $\chi_i > \chi_j$ implies $\alpha_i \leq \alpha_j$ and $\phi_i \leq \phi_j$.*

Proof. See Appendix A.10. □

In line with intuition, spreading a worker who is a poor multitasker but excels at concentrating on a single project across multiple project teams is not optimal. The principal keeps him specialized on the top-ranked projects in the optimum. The opposite is true for workers who are skilled at multitasking but not at concentrating. The least valuable projects are staffed only by workers with the lowest specialization indices. Specialists, those who are assigned to few projects, work

together on large project teams, while generalists may be the only ones employed at a particular project. Holding one cost dimension fixed isolates the effect of the other on assignments and full utilization cutoffs.

Corollary 1. *The following statements follow directly from [Theorem 2](#).*

1. *If workers have homogeneous specialization costs $c_i \equiv c$, then $r_i > r_j$ implies $\alpha_i \leq \alpha_j$ and $\phi_i \leq \phi_j$.*
2. *If workers have homogeneous multitasking costs $r_i \equiv r$, then $c_i > c_j$ implies $\alpha_i \geq \alpha_j$ and $\phi_i \geq \phi_j$.*

While the extensive margin in assignments and utilization is ordered, the intensive margin is not. Effort levels are sensitive to the model parameters. On the one hand, a high multitasking cost leads the principal to reduce that worker's efforts across multiple projects, holding the set of assigned projects fixed. On the other hand, the principal seeks to assign such a worker to a *smaller* set of assigned projects, freeing up his capacity to work more. The net effect can go either way.

Example 1. There are two workers i and j and two projects p and q . The model parameters are $\beta = 1$, $c = 0.6$, $g_{ij} = 0.05$ and multitasking costs $r_i = 0.15$ and $r_j = 0.05$. Worker i is the specialist and worker j is the generalist. The principal has a log-linear Cobb-Douglas valuation. I increase the principal's γ weight on project p from 6 to 7 while keeping her weight on project q at 1.

	$\gamma^{(p)} = 6$			$\gamma^{(p)} = 7$		
	p	q	Total	p	q	Total
Worker i (specialist, $r_i = 0.15$)	1.662	0.577 [†]	2.239	1.724	0.329 [†]	2.053
Worker j (generalist, $r_j = 0.05$)	1.674	1.575	3.249	1.681	1.554	3.235
Gap $e_j^p - e_i^p$	+0.012			−0.044		

[†] Underutilized. Worker i is capped at project q ; worker j is fully utilized everywhere.

Table 5

The intensive margin is not ordered while the extensive margin is unchanged across the two columns. The generalist is fully utilized at both projects, the specialist only at p , yet the ranking of effort levels at p reverses. When $\gamma^{(p)} = 6$, the

generalist works more on p ; when $\gamma^{(p)} = 7$, the specialist does. The assignment pattern stays the same. The total effort column reveals why this is true. When the principal is more biased towards project p , she tightens the specialist's cap at project q . His total effort falls, dragging his marginal cost down, which frees up his capacity to work more on project p . Since the generalist is fully utilized at both projects, the principal has less room to move his efforts around. Whether a specialist works more or less than a generalist on a shared project depends on the levels of the parameters, and not the inherent structure of the assignment.

5.2 The Scope Index

I assume the principal assigns every worker to every project in the optimum and every attribute is heterogeneous, complementarity included. The goal is to examine how workers' full utilization cutoffs are sorted. A natural conjecture is that central workers should be fully utilized at many projects. Because they collaborate well with other workers, allowing them to contribute as much as they wish at many projects amplifies efforts throughout the entire organization. However, doing so backfires because multitasking weakens their contributions, causing other workers to pull back due to a loss in complementarity. This trade-off is captured in a worker's *scope index*, which compares two Katz-Bonacich centralities using two previously defined matrices: the aggregate interaction matrix $\mathbf{P}$ and the team interaction matrix $\mathbf{M}$.

Definition 7. The **scope index** of worker i is $\rho_i := \frac{(\mathbf{P}^{-1}\mathbf{1}_n)_i}{(\mathbf{M}^{-1}\mathbf{1}_n)_i}$.

Because both $\mathbf{P}^{-1}$ and $\mathbf{M}^{-1}$ are nonnegative under [Assumption 1](#), the scope index is also nonnegative. Recall the matrix definitions

$$\mathbf{P} := \mathbf{C} + (m - 1)\mathbf{R} - \mathbf{G} \quad \mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$$

The crucial difference between them lies in their diagonals: $\mathbf{M}$ has $c_i - r_i$ while $\mathbf{P}$ has $c_i + (m - 1)r_i$. In particular, the sign of the multitasking cost makes the numerator and denominator behave differently.

The numerator, which is $(\mathbf{P}^{-1}\mathbf{1}_n)_i$, measures the aggregate increase in total effort across all workers resulting from a unit increase in worker i 's preference for

working on all projects. Notice that the marginal cost of total effort, $c_i + (m - 1)r_i$, grows with multitasking costs. Therefore, workers with low r_i , along with others with strong links to them either directly or through a chain of peers, generate a larger aggregate effort response reflected in a higher $(\mathbf{P}^{-1}\mathbf{1}_n)_i$ value.

Being skilled at multitasking has the opposite effect on the denominator. The entry $(\mathbf{M}^{-1}\mathbf{1}_n)_i$ measures the total increase in effort across all workers on a project from a unit increase in worker i 's preference *holding their total efforts fixed*. Having a low multitasking cost means that it is more expensive to redirect efforts towards project p , which has a net marginal cost of $c_i - r_i$. These workers, along with their collaborators, therefore generate a smaller effort response within a project and have a lower $(\mathbf{M}^{-1}\mathbf{1}_n)_i$ value.

Put together, the scope index measures the advantage a worker has as a generalist versus a specialist. Workers with a high scope index, those with relatively high $(\mathbf{P}^{-1}\mathbf{1}_n)_i$ and low $(\mathbf{M}^{-1}\mathbf{1}_n)_i$ values, are better generalists. It is more beneficial to fully utilize them on a wide range of projects rather than specialize them on a handful of projects. The converse is true for workers with low scope indices, as formalized in the following result.

Theorem 3. *In an optimal project assignment where every worker is assigned to every project, the full utilization cutoffs are sorted such that $\rho_i > \rho_j$ implies $\phi_i \geq \phi_j$.*

Proof. See [Section A.11](#). □

Once the assumption that every worker is assigned to every project is dropped, the sorting index becomes endogenous. Two sets of multipliers must be solved jointly: (1) those on the incentive constraints, which govern full utilization and (2) those on the nonnegativity constraints, which govern assignment. Under heterogeneous complementarity (even under homogeneous costs), neither admits a closed form. Appendix [D](#) develops the resulting index.

6 Comparative Statics

That the incentive constraints are linear in efforts permits us to interpret them as linear budgets over workers' efforts. The comparative statics admit a Slutsky de-

composition into a substitution and income effect. The substitution effect accounts for how the principal substitutes one worker's effort at a project for another if the former becomes more costly to use. The income effect further decomposes into two separate network effects.

To isolate the income effect, I examine the comparative statics of optimal project assignments exhibiting full utilization everywhere. That is, a worker's incentive constraint binds at a project whenever he is assigned there, and I show that these assignments have zero substitution effect. Mathematically, differentiating the linear system in the incentive constraints alone delivers the comparative statics. The income effect separates into a *within-project spillover* and a *network reallocation effect*. The former operates through effort complementarity and is determined by the team interaction matrix $\mathbf{M}$, while the latter reflects how multitasking induces workers to redistribute effort across projects, as dictated by the aggregate interaction matrix $\mathbf{P}$.

6.1 A Slutsky Decomposition

The comparative statics assume that given a shock, assignment and full utilization patterns do not change in the optimum. That is, I focus on a *local* analysis holding fixed the **assignment** and **full utilization sets** $A := \{(i, p) \mid e_i^p > 0\}$ and $\Phi := \{(i, p) \mid i \text{ is fully utilized at } p\}$. I interpret a binding incentive constraint as a linear effort budget.

$$\beta_i^p = c_i e_i^p + r_i \sum_{q \neq p} e_i^q - \sum_{j \neq i} g_{ij} e_j^p$$

The budget reading is as follows: β_i^p is the income of effort available to the principal, c_i is the own price of worker i 's effort at project p , r_i is the price of his effort at other projects and $-g_{ij}$ is a discount because worker j 's effort makes it cheaper for worker i to work more.

I introduce the *substitution* and *income matrices*. Recall that the worker interaction matrix $\mathbf{W}$ encodes each incentive constraint as a row; the relevant submatrix is $\mathbf{W}_\Phi$ with rows restricted to worker-project pairs that are fully utilized. At the

optimum, the efforts and multipliers $(\mathbf{e}_A, \boldsymbol{\lambda}_\Phi)$ satisfy the Kuhn-Tucker conditions:

$$\begin{aligned}\nabla_{\mathbf{e}} V(\mathbf{e}_A) - \mathbf{W}_\Phi^\top \boldsymbol{\lambda}_\Phi &= \mathbf{0} \\ \beta_\Phi - \mathbf{W}_\Phi \mathbf{e}_A &= 0\end{aligned}$$

Implicit differentiation of this system yields the following block Jacobian matrix.

Definition 8. The Jacobian matrix $\mathbf{J}$ and its inverse (if it exists) written in corresponding block matrix form are

$$\mathbf{J} := \begin{bmatrix} \nabla_{\mathbf{e}}^2 V(\mathbf{e}_A) & -\mathbf{W}_\Phi^\top \\ -\mathbf{W}_\Phi & \mathbf{0} \end{bmatrix} \quad \mathbf{J}^{-1} := \begin{bmatrix} (\mathbf{J}^{-1})_{11} & (\mathbf{J}^{-1})_{12} \\ (\mathbf{J}^{-1})_{21} & (\mathbf{J}^{-1})_{22} \end{bmatrix}$$

The **substitution matrix** is $\mathbf{S} := (\mathbf{J}^{-1})_{11}$ and the **income matrix** is $\mathbf{Y} := (\mathbf{J}^{-1})_{12}$.

The substitution matrix $\mathbf{S}$ collects every direction along which the principal can reallocate effort while holding her “purchasing power” of effort fixed. It satisfies $\mathbf{W}_\Phi \mathbf{S} = \mathbf{0}$, and so any reallocation along $\mathbf{S}$ leaves every binding incentive constraint binding as before. This is the effort analog to a movement along a budget line. Moreover, $\mathbf{S}$ is symmetric and negative semidefinite, the latter of which reflects the law of compensated demand.

The income matrix $\mathbf{Y}$ accounts for directions that move the constraints themselves. It satisfies $\mathbf{W}_\Phi \mathbf{Y} = \mathbf{I}$. Each of its columns corresponds to a direction that shifts a binding constraint by exactly one unit and keeps every other binding constraint undisturbed. This is the effort analog of a parallel shift of the budget line after income changes, holding relative prices fixed.

The Slutsky decomposition requires a technical condition to ensure the inverse of the Jacobian $\mathbf{J}$ exists.

Theorem 4. If $\nabla_{\mathbf{e}}^2 V(\mathbf{e}_A)$ is strictly negative definite on $\ker(\mathbf{W}_\Phi^\top)$, then $\mathbf{J}^{-1}$ exists and at the optimum $(\mathbf{e}_A, \boldsymbol{\lambda}_\Phi)$ and for every parameter θ ,

$$\frac{\partial \mathbf{e}_A}{\partial \theta} = \underbrace{\mathbf{S} \left(\frac{\partial \mathbf{W}_\Phi^\top}{\partial \theta} \boldsymbol{\lambda}_\Phi \right)}_{\text{substitution}} + \underbrace{\mathbf{Y} \left(\frac{\partial \beta_\Phi}{\partial \theta} - \frac{\partial \mathbf{W}_\Phi}{\partial \theta} \mathbf{e}_A \right)}_{\text{income}}$$

Proof. See Appendix [A.13](#). □

The condition is sufficient for the optimum to be unique in individual efforts so that the matrix $\mathbf{J}$ is full rank. While $\mathbf{S}$ and $\mathbf{Y}$ represent the directions of the substitution and income effects to which the principal responds, the magnitude of each response is captured by how a shock affects the incentive constraints through $\frac{\partial \mathbf{W}_\Phi}{\partial \theta}$ (and its transpose). To simplify notation, I derive the formulas for $\frac{\partial \mathbf{W}}{\partial \theta}$ which I call the **impact vector**; the corresponding expressions for $\mathbf{W}_\Phi$ are the rows of the following matrices restricted to indices in Φ .

$$\begin{aligned} \frac{\partial \mathbf{W}}{\partial \beta_i^p} &= \mathbf{0} & \frac{\partial \mathbf{W}}{\partial c_i} &= (\boldsymbol{\epsilon}_i \boldsymbol{\epsilon}_i^\top) \otimes \mathbf{I}_m \\ \frac{\partial \mathbf{W}}{\partial g_{ij}} &= -(\boldsymbol{\epsilon}_i \boldsymbol{\epsilon}_j^\top + \boldsymbol{\epsilon}_j \boldsymbol{\epsilon}_i^\top) \otimes \mathbf{I}_m & \frac{\partial \mathbf{W}}{\partial r_i} &= (\boldsymbol{\epsilon}_i \boldsymbol{\epsilon}_i^\top) \otimes (\mathbf{1}_m \mathbf{1}_m^\top - \mathbf{I}_m) \end{aligned}$$

where $\boldsymbol{\epsilon}_i$ is the i -th basis vector.

To see how this Slutsky decomposition compares to that of traditional consumer theory, consider an increase in specialization cost for worker i . Recall ϕ_i denotes the last project worker i is fully utilized at.

$$\frac{\partial \mathbf{e}_A}{\partial c_i} = \sum_{p=1}^{\phi_i} \left(\mathbf{S}_{\cdot, (i,p)} \lambda_i^p - \mathbf{Y}_{\cdot, (i,p)} e_i^p \right)$$

The principal is the consumer, but she faces a separate budget per binding incentive constraint rather than a single one. A price change is usually tracked in one budget, and the income effect is weighted by the quantity of the good whose price moved. Here, c_i is the price of worker i 's effort at every project he is fully utilized, and a change in c_i moves all ϕ_i budget lines at the same time.

6.2 Decomposing the Income Effect

There is a class of project assignments with no substitution effect. When workers are fully utilized whenever they are assigned, any reallocation of efforts along one incentive constraint violates other incentive constraints. Geometrically, the feasible set, which is defined by linear constraints, forms a polytope. Underuti-

lization places the optimum on a face of the polytope and the substitution effect corresponds to movements across that surface.

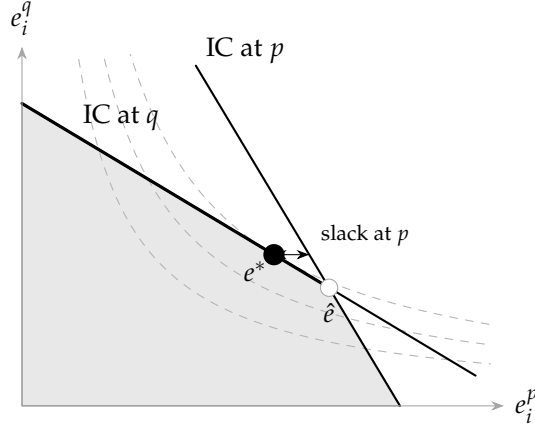


Figure 4: Underutilization at $\mathbf{e}^*$ for one worker and two projects.

If the optimum lands on a vertex of the polytope ($\hat{\mathbf{e}}$ in Figure 4), movement along an incentive constraint necessarily moves off another binding constraint. These are project assignments with no underutilization.

Proposition 7. *If every worker assigned to a project is fully utilized in the optimum, then $\mathbf{S} = \mathbf{0}$.*

Proof. See Appendix A.12. □

By isolating the income effect in projects exhibiting full utilization, I demonstrate that it can be further decomposed into two separate components: (1) a *within-project spillover* where workers redirect effort towards a project receiving a positive shock and (2) a *network reallocation effect*, where workers adjust effort across different projects to manage multitasking costs.

If the principal assigns and fully utilizes every worker at every project, then the assignment is given by the no-intervention outcome $\hat{\mathbf{e}} = \mathbf{W}^{-1}\boldsymbol{\beta}$. The income matrix is $\mathbf{W}^{-1}$.¹⁷ To compute the comparative statics at the project-level, I define project-specific impact vectors.

¹⁷Reading off Theorem 4, we have $\frac{\partial \mathbf{e}}{\partial \theta} = \mathbf{W}^{-1} \left(\frac{\partial \boldsymbol{\beta}}{\partial \theta} - \frac{\partial \mathbf{W}}{\partial \theta} \mathbf{e} \right)$.

Definition 9. For parameter $\theta \in \{\beta_j^q, c_j, r_j, g_{jk}\}$, $\mathbf{d}^p(\theta) \in \mathbb{R}^n$ is the **impact vector at project p** and is given by

$$\begin{aligned} \mathbf{d}^p(\beta_j^q) &= \delta_{pq} \boldsymbol{\epsilon}_j & \mathbf{d}^p(c_j) &= -e_j^{p*} \boldsymbol{\epsilon}_j \\ \mathbf{d}^p(g_{jk}) &= e_k^{p*} \boldsymbol{\epsilon}_j + e_j^{p*} \boldsymbol{\epsilon}_k & \mathbf{d}^p(r_j) &= -(E_j^* - e_j^{p*}) \boldsymbol{\epsilon}_j \end{aligned}$$

where δ_{pq} is the Kronecker delta, $\boldsymbol{\epsilon}_j$ is the j -th standard basis vector and E_j is worker j 's total effort.

Splitting the income effect into the two network effects requires a matrix decomposition of the worker interaction matrix into the familiar matrices $\mathbf{M}^{-1}$ and $\mathbf{P}^{-1}$.

Proposition 8. *The following decomposition holds*

$$\mathbf{W}^{-1} = \mathbf{M}^{-1} \otimes \mathbf{I}_m - (\mathbf{M}^{-1} \mathbf{R} \mathbf{P}^{-1}) \otimes \mathbf{1}_m \mathbf{1}_m^\top.$$

This implies that in an optimal project assignment with full utilization and assignment, the following holds for every parameter θ and every project p

$$\frac{\partial \mathbf{e}^p}{\partial \theta} = \underbrace{\mathbf{M}^{-1}(\mathbf{d}^p(\theta))}_{\text{within-project spillover}} - \underbrace{\mathbf{M}^{-1} \mathbf{R} \mathbf{P}^{-1} \sum_{q=1}^m \mathbf{d}^q(\theta)}_{\text{network reallocation}}$$

Moreover $\frac{\partial e_i^p}{\partial \beta_j^p} > 0$ for all j and $\frac{\partial e_i^p}{\partial \beta_j^q} < 0$ for all $q \neq p$.

Proof. See Appendix A.14. □

To see the two network effects in action, I focus on how the principal adjusts worker i 's effort at project p given a positive preference shock for worker j at project q .

$$\frac{\partial e_i^p}{\partial \beta_j^q} = \underbrace{\mathbf{M}_{ij}^{-1} \delta_{pq}}_{\text{within-project spillover}} - \underbrace{\sum_{\ell=1}^n \mathbf{M}_{i\ell}^{-1} r_\ell \mathbf{P}_{\ell j}^{-1}}_{\text{network reallocation}}$$

The first term is local to the project experiencing the shock. Worker j is willing to work more there, and complementarity makes everyone else supply more effort. The $\mathbf{M}_{ij}^{-1}$ entry counts the total complementarity effect for worker i at project q while total efforts are held fixed.

The second term represents a separate reallocation channel on the entire network. The $\mathbf{P}_{\ell j}^{-1}$ entry captures how much worker ℓ 's total effort rises in response to an increase in worker j 's preference for working on any project. Multiplying by r_ℓ converts that extra effort into an increase in worker ℓ 's marginal cost, which applies at every project he works on. In response, he reduces effort at every project, and $\mathbf{M}_{i\ell}^{-1}$ propagates this back through the within-project network to worker i : he experiences a loss in the complementarity from worker ℓ . The network reallocation effect causes a drop in quality for every project not subject to the positive preference shock.

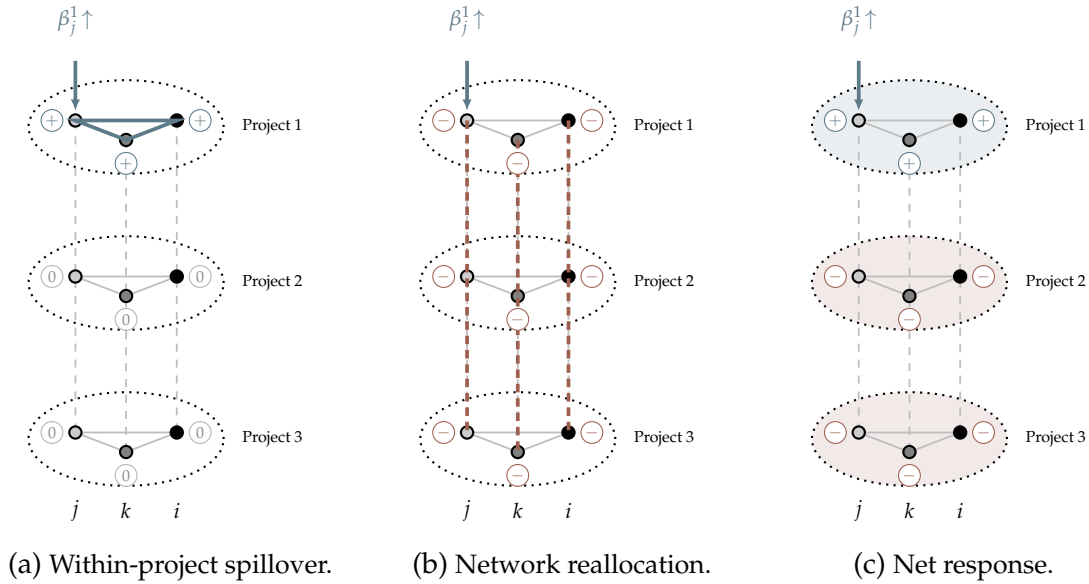
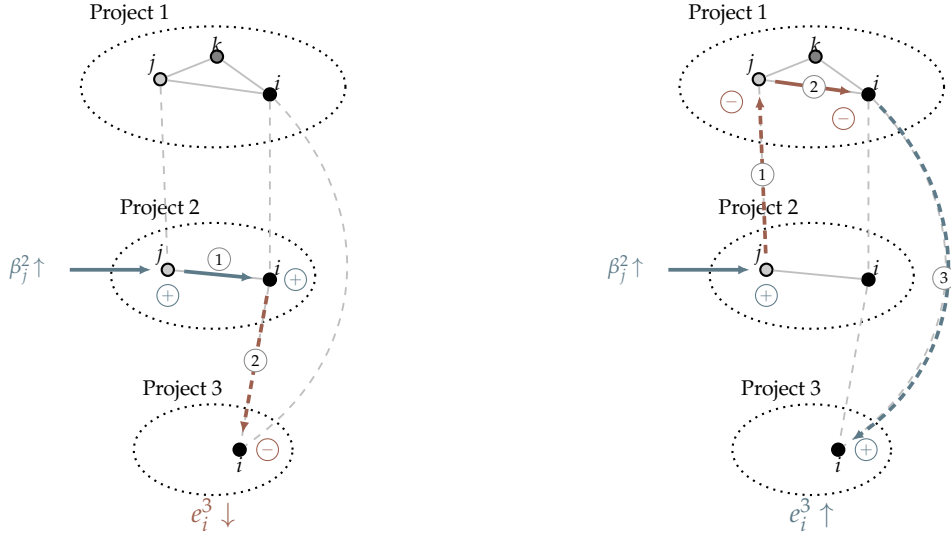


Figure 5: Nodes of the same color correspond to the same worker. The ellipses represent different projects. Worker j receives a preference shock at project 1.

What changes when the principal fully utilizes workers but does not assign them to every project? The magnitude and direction of a shock transmission becomes influenced by the pattern of active project assignments. The following example illustrates.

Example 2. Suppose there are three projects (1, 2, 3) and three workers (i, j, k), and the optimal project assignment has the following nested structure: worker i is assigned to all 3 projects, worker j is assigned to projects 1 and 2 and worker k is assigned to project 1.



(a) The direct route $(j, 2) \rightarrow (i, 2) \rightarrow (i, 3)$. Complementarity raises worker i 's effort at project 2; his multitasking cost then pulls him off project 3.

(b) The indirect route, $(j, 2) \rightarrow (j, 1) \rightarrow (i, 1) \rightarrow (i, 3)$. Worker j reduces effort at project 1, worker i loses complementarity, and he reduces effort at project 1. This frees up effort for i to spend at project 3.

Figure 6: Two routes for the same shock under partial assignment.

Consider an increase in worker j 's preference for working on project 2, denoted by $\beta_j^{(2)}$. This shock propagates through two distinct channels. Worker i raises his efforts at project 2 due to an increase in complementarity, and cuts back on his efforts at project 3. The other channel sees worker j withdraw from project 1. Worker i follows suit at project 1, saving capacity to work on his other projects. This effect pushes worker i to *raise* his effort at project 3. The net direction of the shock is ambiguous and hinges on the strength of his effort complementarity. For example, if g_{ij} is large, then his gain in effort at project 2 dominates, leading to higher total effort and an overall contraction at project 3. On the other hand, if g_{ij} is low, worker i 's effort reduction at project 1 could dominate and he works more on project 3. In contrast to the full assignment case, a positive preference shock for one project could *raise* the quality of another project.

Under partial assignment, the project impact vectors remain unchanged. The primary distinction lies in the team interaction matrix. Within-project spillovers for project p are governed by $\mathbf{M}^{(A^p)}$, the principal submatrix of $\mathbf{M}$ restricted to workers assigned to p . Likewise, the network reallocation effect changes as well

because each project carries a different $\mathbf{M}^{(A^p)}$ matrix. The operator that captures how total efforts respond to the shock is no longer $\mathbf{P}$, but $\mathcal{P}$, which depends on the pattern of assignment. It has entries

$$\mathcal{P}_{ij} = \delta_{ij} + r_j \sum_{p \in P_i \cap P_j} (\mathbf{M}^{(A^p)})_{ij}^{-1}.$$

where P_i is the set of projects worker i is assigned to.

Proposition 9. *In an optimal project assignment with full utilization and partial assignment, the following holds for every parameter θ and every project p*

$$\frac{\partial e^p}{\partial \theta} = \underbrace{(\mathbf{M}^{(A^p)})^{-1} \mathbf{d}^p(\theta)}_{\text{within-project spillover}} - \underbrace{(\mathbf{M}^{(A^p)})^{-1} \mathbf{R}(\mathcal{P}^{-1} \mathbf{v}(\theta))}_{\text{network reallocation}}$$

where the right-hand vector $\mathbf{v}$ has entries

$$\mathbf{v}_i(\theta) := \sum_{p \mid i \in A^p} ((\mathbf{M}^{(A^p)})^{-1} \mathbf{d}^p(\theta))_i$$

Proof. See Appendix A.15. □

The comparative statics are carried by the assignment-specific interaction matrices.¹⁸ As the example demonstrated, a positive preference shock could raise the qualities of other projects; imbalanced project assignments lead to different routes the shock can take.

7 Discussion

7.1 Effort Caps and Contracts

By studying optimal project assignments via effort caps in isolation, I prohibit the principal from contracting on effort. A natural question is whether she could

¹⁸One can readily verify that Proposition 8 is a special case of the above: under full assignment $(\mathbf{M}^{(A^p)})^{-1} \equiv \mathbf{M}^{-1}$, so $\mathcal{P} = \mathbf{I} + m\mathbf{M}^{-1}\mathbf{R} = \mathbf{M}^{-1}\mathbf{P}$ and $\mathbf{v}(\theta) = \mathbf{M}^{-1} \sum_{q=1}^m \mathbf{d}^q(\theta)$.

achieve the outcomes if transfers were allowed. The comparison here is about implementable outcomes, not which instrument the principal would prefer as the main analysis concerns environments where contracting is unavailable. Answering the latter requires additional assumptions on how her valuation of projects relates to the costs of paying her workers.

I show that effort caps implement effort profiles that contracts which reward effort cannot.

Proposition 10. *Suppose worker i 's payoff under contract w_i is given by $U_i(\mathbf{e}_i, \mathbf{e}_{-i}) + w_i(\mathbf{e}_i, \mathbf{e}_{-i})$ where w_i is nondecreasing in each e_i^p . No effort profile where worker i is underutilized at project p is an equilibrium.*

Proof. See Appendix A.16. □

Equivalently, any contract implementing a profile with underutilization for worker i at project p must be strictly decreasing in e_i^p . For an illustration, consider a linear contract for each worker i

$$w_i(\mathbf{e}_i) = \sum_{p=1}^m b_i^p e_i^p \text{ where } b_i^p \in \mathbb{R}.^{19}$$

To implement an effort profile, the required rate is the negative of the worker's marginal utility at the target. To underutilize a worker and leave his incentive constraint slack, the principal must fine him for working. Effort caps and *nonnegative* wages are different instruments that move effort in the opposite direction. A cap holds a worker back on a project he would like to contribute more to, while a positive wage pushes him to do so. Instead, contracts can achieve effort profiles where worker marginal utilities are negative, something effort caps cannot accomplish.

¹⁹I leave out a fixed payment because it does not affect workers' incentives to exert effort. It does, however, affect whether limited liability binds and can be chosen so that total pay is always nonnegative.

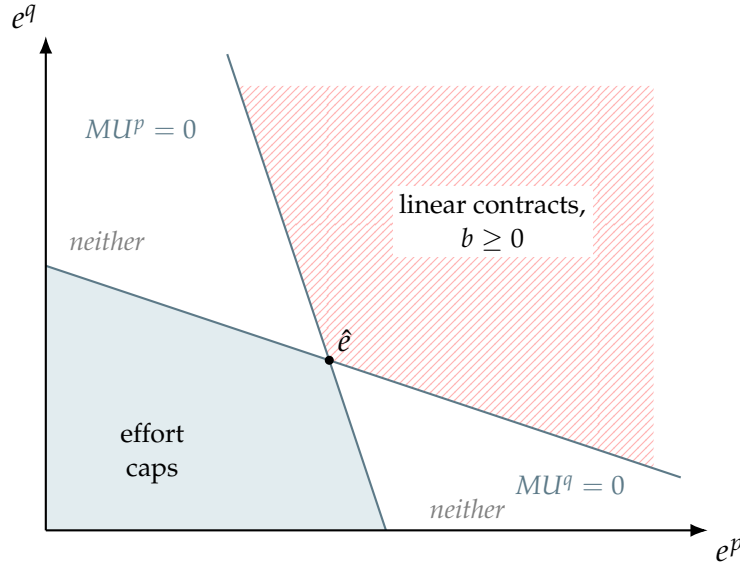


Figure 7: Effort profiles implementable by caps and by contracts, for one worker and two projects. The lines are the worker's incentive constraints. The two sets meet only at the unconstrained equilibrium $\hat{e}$, where every marginal utility vanishes. In the two unshaded regions, neither instrument alone suffices.

7.2 Asymmetric Complementarities

I imposed symmetric complementarities early on to write the workers' effort game as an exact potential game, which buys us uniqueness of equilibrium. Beyond that, complementarities can be asymmetric, and the main structural results still hold.

It is imperative that worker interactions are complements and not substitutes within a project. If interactions are substitutes i.e. $g_{ij} < 0$, then the team-coordinated effort transfer argument is no longer available. Shifting a worker's effort from a low-value project to a high-value one encourages other workers to desire more work at the former, rendering the arbitrage moot. The nested hierarchy structure no longer holds.

7.3 Effort Budgets

Suppose there is a maximum level of worker effort that the principal can implement in equilibrium e.g. a 40-hour work week. The restriction places an additional

constraint in the principal’s convex optimization problem where, for all workers i , $\sum_{p=1}^m e_i^p \leq T_i$, where T_i is an exogenously given hard limit. The main structural results still hold because the team-coordinated transfer keeps workers’ total efforts constant. However, the sorting results are not robust. For example, even if a worker has a high scope index, implying they should be fully utilized across many project teams, they may operate with a very low T_i . Consequently, the principal cannot fully utilize them in as many teams.

8 Concluding Remarks

Organizations do not pay for effort on a day-to-day basis. Instead, they choose how much they want implemented by assignments. Effort caps are not a substitute for incentives and deserve their own analysis. They are essentially two instruments in one: they select which workers collaborate with whom, and dictate the scope of what a worker works on in the organization. Optimally assigning workers to projects by setting effort caps turns out to have a much more structured solution than the richness of the model suggests. A single endogenous ranking of projects sorts every worker’s assignment into three consecutive intervals— fully utilized, assigned but underutilized, unassigned. I showed that centrality alone does not sort specialists from generalists. Instead, a ratio of two Katz-Bonacich centralities sorts workers by weighing the value of strong ties within a team and against the cost of spreading it across many project teams. Because the assignment selects the network of interactions, an organization’s response to shocks is implicitly designed by the principal. A shock favorable to one project could end up being bad news for other projects in the organization, depending on the pattern of assignment and utilization in the optimum.

There are many promising extensions for future research. Because this paper studies assignment in isolation, a natural next step is to study the joint design of assignments and incentives, especially given that they affect effort in opposite ways. Project assignments could also be studied in tandem with career concerns, as success on certain projects could advance a worker’s career more than others. Lastly, [Assumption 1](#) guarantees a unique equilibrium and nullifies equilibrium selection concerns. Relaxing it allows us to study which project assignments remain robust

when the worst equilibrium is played. I hope that this work inspires future research in studying staffing decisions and project assignment through the lens of a network design problem.

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A Omitted Proofs

A.1 Proof of Proposition 1

Worker i maximizes $U_i(\mathbf{e}_i, \mathbf{e}_{-i})$, which is concave in own efforts, over $\prod_{p=1}^m [0, \bar{e}_i^p]$. Therefore the first-order conditions are necessary and sufficient. His effort $\mathbf{e}_i$ is a best response if and only if for every project p the following holds:

- (1) If $0 < e_i^p < \bar{e}_i^p$, then $\frac{\partial U_i}{\partial e_i^p} = 0$.
- (2) If $e_i^p = \bar{e}_i^p > 0$, then $\frac{\partial U_i}{\partial e_i^p} \geq 0$.
- (3) If $e_i^p = 0 < \bar{e}_i^p$, then $\frac{\partial U_i}{\partial e_i^p} \leq 0$.
- (4) If $e_i^p = \bar{e}_i^p = 0$, then no restriction on $\frac{\partial U_i}{\partial e_i^p}$ is required.

(Forward Direction) Suppose $\mathbf{e}$ is a Nash equilibrium given $\bar{\mathbf{e}}$ and assume that $e_i^p > 0$. Since $\bar{e}_i^p \geq e_i^p > 0$, then either Condition (1) or (2) holds. Compute

$$\frac{\partial U_i}{\partial e_i^p} = \beta_i^p + \sum_{j \neq i} g_{ij} e_j^p - c_i e_i^p - r_i \sum_{q \neq p} e_i^q \geq 0 \implies \beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q$$

which is the desired inequality.

(Reverse Direction). Let $\mathbf{e}$ satisfy $\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q$ whenever $e_i^p > 0$ and set $\bar{\mathbf{e}} = \mathbf{e}$. Fix a worker i and a project p . If $e_i^p > 0$, then $e_i^p = \bar{e}_i^p > 0$ and Condition (2) is satisfied because $\frac{\partial U_i}{\partial e_i^p} \geq 0$ by assumption. If $e_i^p = 0$, then $\bar{e}_i^p = 0$ and Condition (4) is vacuously satisfied. The worker has no deviation available is forced to exert zero effort. Every project satisfies the best response conditions, so $\mathbf{e}_i$ is a best response to $\mathbf{e}_{-i}$ and $\mathbf{e}$ is a Nash equilibrium. $\square$

A.2 Proof of Proposition 2

I first show the following lemma.

Lemma 3. Under Assumption 1, the following statements are true.

1. The inverse of $\mathbf{P}$ exists and is nonnegative.
2. Every effort profile satisfying every incentive constraint satisfies $\mathbf{E} \leq \tilde{\mathbf{E}} := \mathbf{P}^{-1} \sum_{p=1}^m \boldsymbol{\beta}^p$.

Proof. [Assumption 1](#) and complementarity between workers guarantee $\mathbf{P}$ has a nonnegative inverse matrix $\mathbf{P}^{-1} \geq \mathbf{0}$. This is because $\mathbf{P}$ is positive definite: for any $\mathbf{u} \neq \mathbf{0}$, $m\mathbf{u}^\top \mathbf{P}\mathbf{u} = (\mathbf{u} \otimes \mathbf{1}_m)^\top \mathbf{W}(\mathbf{u} \otimes \mathbf{1}_m) > 0$. It also has nonpositive off-diagonals $-g_{ij}$, and so it is a symmetric M -matrix. Having a nonnegative inverse matrix is a property of M -matrices.

Suppose an effort profile satisfies every incentive constraint. Fix a project p and stack each worker's incentive constraint:

$$\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q \implies \beta^p \geq (\mathbf{C} - \mathbf{R} - \mathbf{G})\mathbf{e}^p + \mathbf{R}\mathbf{E}$$

where $\mathbf{E} := [\sum_{p=1}^m e_1^p \cdots \sum_{p=1}^m e_n^p]$ is the vector of worker total efforts. The implication holds by rewriting $\sum_{q \neq p} e_i^q = \sum_{q=1}^m e_i^q - e_i^p$. Finally, aggregate the incentive constraints across all m projects to obtain an inequality in total efforts

$$\sum_{p=1}^m \beta^p \geq (\mathbf{C} - \mathbf{R} - \mathbf{G})\mathbf{E} + m\mathbf{R}\mathbf{E} \implies \sum_{p=1}^m \beta^p \geq \mathbf{P}\mathbf{E}.$$

Therefore $\tilde{\mathbf{E}} := \mathbf{P}^{-1} \sum_{p=1}^m \boldsymbol{\beta}^p$ is the upper bound on the total efforts. $\square$

Continuing with the main proof, assume that $r_i \tilde{E}_i < \min_p \beta_i^p$. By construction, $\tilde{E}_i$ satisfies

$$(c_i + r_i(m-1))\tilde{E}_i = \sum_{p=1}^m \beta_i^p + \sum_{j \neq i} g_{ij} \tilde{E}_j$$

Let $\mathcal{A}_i^{(k)}$ be the set of the k largest entries of $\{\beta_i^p\}_{p=1}^m$. Subtract to obtain

$$(c_i + r_i(m-1))\tilde{E}_i - \sum_{p \in \mathcal{A}_i^{(k)}} \beta_i^p - \sum_{j \neq i} g_{ij} \tilde{E}_j = \sum_{p \notin \mathcal{A}_i^{(k)}} \beta_i^p \geq (m-k) \min_p \beta_i^p \geq (m-k)r_i \tilde{E}_i$$

Rearrange to get

$$(m-k)r_i \tilde{E}_i \leq (c_i + r_i(m-1))\tilde{E}_i - \sum_{p \in \mathcal{A}_i^{(k)}} \beta_i^p - \sum_{j \neq i} g_{ij} \tilde{E}_j$$

$$\implies (c_i + r_i(k-1))\tilde{E}_i \geq \sum_{p \in \mathcal{A}_i^{(k)}} \beta_i^p + \sum_{j \neq i} g_{ij}\tilde{E}_j \quad (*)$$

Now I show that for any implementable $\mathbf{e}$, the total efforts $E_i := \sum_{p=1}^m e_i^p$ are bounded from above by $\tilde{E}_i$. Summing worker i 's incentive constraints over projects $p \in P_i$ gives

$$\beta_i^p + \sum_{j \neq i} g_{ij}e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q \implies \sum_{p \in P_i} \beta_i^p + \sum_{j \neq i} g_{ij} \sum_{p \in P_i} e_j^p \geq \sum_{p \in P_i} (c_i e_i^p + r_i \sum_{q \neq p} e_i^q)$$

The cost side simplifies to

$$\sum_{p \in P_i} (c_i e_i^p + r_i \sum_{q \neq p} e_i^q) = (c_i + r_i(|P_i| - 1))E_i$$

Now suppose that $\zeta := \max_i (E_i / \tilde{E}_i) > 1$. Then multiply and divide by $\tilde{E}_i$ on both the left and right sides to get

$$\zeta(c_i + r_i(|P_i| - 1))\tilde{E}_i \leq \sum_{p \in P_i} \beta_i^p + \zeta \sum_{j \neq i} g_{ij}\tilde{E}_j = \sum_{p \in P_i} \beta_i^p + \sum_{j \neq i} g_{ij}\tilde{E}_j + (\zeta - 1) \sum_{j \neq i} g_{ij}\tilde{E}_j$$

Applying the $(*)$ relation, we have

$$\sum_{p \in P_i} \beta_i^p + \sum_{j \neq i} g_{ij}\tilde{E}_j + (\zeta - 1) \sum_{j \neq i} g_{ij}\tilde{E}_j \leq (c_i + r_i(|P_i| - 1))\tilde{E}_i + (\zeta - 1) \sum_{j \neq i} g_{ij}\tilde{E}_j$$

Chaining the inequality from the two lines above:

$$(\zeta - 1)(c_i + r_i(|P_i| - 1))\tilde{E}_i \leq (\zeta - 1) \sum_{j \neq i} g_{ij}\tilde{E}_j \implies (c_i + r_i(|P_i| - 1))\tilde{E}_i \leq \sum_{j \neq i} g_{ij}\tilde{E}_j$$

This implies that $\sum_{p \in \mathcal{A}_i^{(k)}} \beta_i^p \leq 0$, which is a direct contradiction that $\beta_i^p > 0$ for all i and p .

Now assume $\mathbf{e}$ is implementable with $e_i^p = 0$. Then $\sum_{q \neq p} e_i^q = E_i \leq \tilde{E}_i$ and so

$$\beta_i^p + \sum_{j \neq i} g_{ij}e_j^p \geq \min_p \beta_i^p > r_i \tilde{E}_i \geq r_i \sum_{q \neq p} e_i^q$$

Because worker i 's incentive constraint at p is always satisfied given zero effort e_i^p ,

every implementable profile must therefore satisfy every incentive constraint and the feasible set of the mixed-integer nonlinear program in Appendix B coincides with that of the convex program. $\square$

A.3 Proof of Theorem 1

Assume that $V^p > V^q$ and that worker i is assigned to project q but underutilized at project p . The team-coordinated effort transfer in Section 4.2 offers a contradiction at the optimum. If $A^q \subset N$, I construct the following transfer:

$$x_j = (\delta(\mathbf{M}^{(A^q)})^{-1}\mathbf{e}_i)_j \text{ for } j \in A^q \text{ and } x_j = 0 \text{ for } j \notin A^q$$

The inverse $(\mathbf{M}^{(A^q)})^{-1}$ exists and is nonnegative because $\mathbf{M}$ is positive definite and has negative off-diagonal entries by Appendix A.4. Therefore $\mathbf{M}^{(A^q)}$ is a M -matrix because it inherits positive definiteness from $\mathbf{M}$ by Sylvester's criterion and has negative off-diagonal entries. M -matrices have nonnegative inverses.

Using this transfer, the principal moves an assigned worker's effort from q to p to precisely offset the total complementarity forgone at q and exploit the total complementarity gained at p . To show why this transfer works, examine how the slacks at project q change, which is where the naïve transfer failed. I use a block matrix computation. Let $\bar{A}^q$ be the set of all workers not assigned to project q . Partition $\mathbf{M}$ into blocks according to whether a worker is assigned to project q . To be clear, $\mathbf{M}^{(A^q),(\bar{A}^q)}$ and $\mathbf{M}^{(\bar{A}^q),(A^q)}$ are submatrices of $\mathbf{M}$ containing complementarities $(-g_{ij})$ between workers assigned to q and those who are not. The change in slacks is

$$\Delta \mathbf{s}^q = \mathbf{M}\mathbf{x} = \begin{bmatrix} \mathbf{M}^{(A^q)} & \mathbf{M}^{(A^q),(\bar{A}^q)} \\ \mathbf{M}^{(\bar{A}^q),(A^q)} & \mathbf{M}^{(\bar{A}^q)} \end{bmatrix} \begin{bmatrix} (x_j)_{j \in A^q} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{M}^{(A^q)}(x_j)_{j \in A^q} \\ \mathbf{M}^{(\bar{A}^q),(A^q)}(x_j)_{j \in A^q} \end{bmatrix}$$

The first $|A^q|$ rows correspond to the incentive constraints of workers on the project team at q , and this is where the transfer $\mathbf{x}$ operates. By construction of x_j , we compute $\mathbf{M}^{(A^q)}(x_j)_{j \in A^q} = \delta\mathbf{e}_i$. Worker j 's slack at q remains exactly the same as before the transfer. His decrease in effort at q matches the loss in complementarity driven by the reduction in his teammates' efforts, where all reductions are calibrated in

the same way. Only worker i 's slack at q changes; it rises by δ . The remaining rows correspond to workers not assigned to q . They lose some slack, but recall that under [Assumption 3](#), incentive constraints do not bind at zero effort. Thus, they have strictly positive slack to absorb the loss given δ is sufficiently small. All of the same observations hold in reverse at project p .

Notice this covers all cases: underutilizing worker i at project p means he can be either assigned or unassigned.

For the case when $V^p = V^q$, see [Proposition 3](#). □

A.4 Proof of [Lemma 1](#)

By [Appendix A.6](#), $\mathbf{M}$ is an M -matrix so $\mathbf{M}^{-1} \geq \mathbf{0}$.

Note that $\mathbf{M} = \mathbf{C} - \mathbf{R} - \mathbf{G}$ is a regular splitting because $\mathbf{G}$ has zero diagonal and positive definiteness gives $(\mathbf{C} - \mathbf{R})_{ii} = c_i - r_i > 0$. Hence $(\mathbf{C} - \mathbf{R})^{-1} \geq \mathbf{0}$ and $\mathbf{G} \geq \mathbf{0}$. Every regular splitting of a nonsingular M -matrix is convergent (Theorem 2.3 of Chapter 6 in [Berman and Plemmons \(1994\)](#)), and therefore the spectral radius $\rho((\mathbf{C} - \mathbf{R})^{-1}\mathbf{G}) < 1$. The Neumann series for $(\mathbf{I} - (\mathbf{C} - \mathbf{R})^{-1}\mathbf{G})^{-1}$ converges, and $\mathbf{M}^{-1} = (\mathbf{I} - (\mathbf{C} - \mathbf{R})^{-1}\mathbf{G})^{-1}(\mathbf{C} - \mathbf{R})^{-1} = \sum_{t=1}^{\infty} ((\mathbf{C} - \mathbf{R})^{-1}\mathbf{G})^{t-1}(\mathbf{C} - \mathbf{R})^{-1}$. □

A.5 Proof of [Proposition 3](#)

Assume that $V^p = V^q$. The stationarity condition for the Lagrangian of the principal's optimization problem with respect to e_i^p is

$$V^p + \sum_{j \neq i} g_{ij} \lambda_j^p - c_i \lambda_i^p - r_i \sum_{q \neq p} \lambda_i^q + \mu_i^p = 0$$

where λ_i^p and μ_i^p are the multipliers on the (i, p) incentive and nonnegativity constraints. Written in terms of worker i 's sum of multipliers on the incentive constraints $\Lambda_i := \sum_{p=1}^m \lambda_i^p$ is:

$$V^p + \sum_{j \neq i} g_{ij} \lambda_j^p + \mu_i^p = (c_i - r_i) \lambda_i^p + r_i \Lambda_i$$

where $\Lambda_i := \sum_{p=1}^m \lambda_i^p$. Subtract the stationarity condition for (i, q) from (i, p) . The multipliers satisfy the following equality

$$\sum_{j \neq i} g_{ij}(\lambda_j^p - \lambda_j^q) + (\mu_i^p - \mu_i^q) = (c_i - r_i)(\lambda_i^p - \lambda_i^q).$$

Stack the equations for each worker

$$\boldsymbol{\mu}^p - \boldsymbol{\mu}^q = \mathbf{M}(\boldsymbol{\lambda}^p - \boldsymbol{\lambda}^q).$$

Suppose the two projects are staffed differently so that $\mathbf{d} := \boldsymbol{\lambda}^p - \boldsymbol{\lambda}^q \neq \mathbf{0}$. Because $\mathbf{M}$ is positive definite by [Assumption 1](#), $\mathbf{d}^\top \mathbf{M} \mathbf{d} > 0$ while the left hand side has $\mathbf{d}^\top (\boldsymbol{\mu}^p - \boldsymbol{\mu}^q) = -(\boldsymbol{\lambda}^p \boldsymbol{\mu}^q - \boldsymbol{\lambda}^q \boldsymbol{\mu}^p) \leq 0$ which is a contradiction. Thus it must be that $\boldsymbol{\lambda}^p = \boldsymbol{\lambda}^q$ and $\boldsymbol{\mu}^p = \boldsymbol{\mu}^q$.

For the second statement, fix a worker i underutilized at projects p and q and without loss of generality let $V^p \geq V^q$. Underutilization requires $\mu_i^p = 0$ and $\lambda_i^p = 0$, and likewise at q . Row i of the stationarity condition therefore loses its own-multiplier term $(c_i - r_i)\lambda_i^p$ and collapses to

$$V^p = r_i \Lambda_i - \sum_{j \neq i} g_{ij} \lambda_j^p$$

and identically at q . Subtracting the two, the term $r_i \Lambda_i$ cancels because it carries no project index:

$$V^p - V^q = \sum_{j \neq i} g_{ij} (\lambda_j^q - \lambda_j^p).$$

Here, I exploit a result on the monotonicity of λ that is established using linear complementarity theory.

Lemma 4. *If $V^p \geq V^q$, then $\boldsymbol{\lambda}^p \geq \boldsymbol{\lambda}^q$.*

Proof. There is a linear complementarity relationship between λ_i^p and μ_i^p .

1. If $\lambda_i^p > 0$, then $e_i^p > 0$ and $\mu_i^p = 0$.
2. If $\mu_i^p > 0$, then $e_i^p = 0$ and $\lambda_i^p = 0$.

Rearrange the stationarity condition for e_i^p :

$$\mu_i^p = (c_i - r_i)\lambda_i^p - \sum_{j \neq i} g_{ij}\lambda_j^p + r_i\Lambda_i - V^p \implies \boldsymbol{\mu}^p = \mathbf{M}\boldsymbol{\lambda}^p + \mathbf{R}\boldsymbol{\Lambda} - V^p\mathbf{1}_n$$

where $\mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$ and $\boldsymbol{\Lambda} = [\Lambda_1, \dots, \Lambda_n]^\top$. Write $\mathbf{w}^p := \mathbf{R}\boldsymbol{\Lambda} - V^p\mathbf{1}_n$. The following are two facts established in linear complementarity theory.

1. Because $\mathbf{M}$ is positive definite, $\text{LCP}(\mathbf{M}, \mathbf{w}^p)$ has a unique solution for each right-hand vector $\mathbf{w}^p$ (Theorem 3.1.6 of [Cottle et al. \(2009\)](#)).
2. Because $\mathbf{M}$ is a Z-matrix, the unique solution $\boldsymbol{\lambda}^*(\mathbf{w}^p)$ to $\text{LCP}(\mathbf{M}, \mathbf{w}^p)$ is the least element of the feasible set (Theorem 3.11.6 of [Cottle et al. \(2009\)](#)).

Clearly $V^p \geq V^q$ implies $\mathbf{w}^p \leq \mathbf{w}^q$ (component-wise). Then $\mathbf{M}\boldsymbol{\lambda} + \mathbf{w}^q \geq \mathbf{M}\boldsymbol{\lambda} + \mathbf{w}^p$. This implies that $\{\boldsymbol{\lambda} \mid \mathbf{M}\boldsymbol{\lambda} + \mathbf{w}^p \geq \mathbf{0}\} \subseteq \{\boldsymbol{\lambda} \mid \mathbf{M}\boldsymbol{\lambda} + \mathbf{w}^q \geq \mathbf{0}\}$. By the least-element property, the solution $\boldsymbol{\lambda}$ must satisfy:

$$\boldsymbol{\lambda}^q = \min_{\boldsymbol{\lambda}} \{\boldsymbol{\lambda} \mid \mathbf{M}\boldsymbol{\lambda} + \mathbf{w}^q \geq \mathbf{0}\} \leq \min_{\boldsymbol{\lambda}} \{\boldsymbol{\lambda} \mid \mathbf{M}\boldsymbol{\lambda} + \mathbf{w}^p \geq \mathbf{0}\} = \boldsymbol{\lambda}^p$$

This establishes monotonicity of $\boldsymbol{\lambda}^p$ with respect to V^p . □

Because $V^p \geq V^q$, we have $\boldsymbol{\lambda}^p \geq \boldsymbol{\lambda}^q$. As $g_{ij} \geq 0$, the right-hand side is nonpositive, so $V^p \leq V^q$ and therefore $V^p = V^q$. Because p and q were arbitrary projects at which worker i is underutilized, $V^{\phi_i+1} = \dots = V^{\alpha_i}$. □

A.6 Proof of [Lemma 2](#)

In $\hat{\mathbf{e}}$, the block corresponding to project p satisfies

$$(\mathbf{C} - \mathbf{R} - \mathbf{G})\hat{\mathbf{e}}^p + \mathbf{R}\hat{\mathbf{E}} = \boldsymbol{\beta}^p$$

where $\hat{\mathbf{E}} = [\sum_{p=1}^m \hat{e}_1^p \cdots \sum_{p=1}^m \hat{e}_n^p]$ is the total effort vector associated with $\hat{\mathbf{e}}$. Summing over all p , we get

$$\underbrace{((\mathbf{C} - \mathbf{R} - \mathbf{G}) + m\mathbf{R})}_{=\mathbf{P}} \hat{\mathbf{E}} = \sum_{p=1}^m \boldsymbol{\beta}^p$$

Therefore $\hat{\mathbf{E}} = \tilde{\mathbf{E}}$ and we have that

$$\hat{\mathbf{e}}^p = (\mathbf{C} - \mathbf{R} - \mathbf{G})^{-1}(\boldsymbol{\beta}^p - \mathbf{R}\tilde{\mathbf{E}})$$

Next, I show that $\mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$ is a positive definite M -matrix and so its inverse is nonnegative and has a strictly positive diagonal. Take $\mathbf{x} \in \mathbb{R}^n$ where $\mathbf{x} \neq \mathbf{0}$ and $\mathbf{y} \in \mathbb{R}^m$ with $\mathbf{y} \neq \mathbf{0}$ where $\mathbf{y} \cdot \mathbf{1}_m = \mathbf{0}$. Then $\mathbf{x}^\top \mathbf{M} \mathbf{x} = (\mathbf{x} \otimes \mathbf{y})^\top \mathbf{W}(\mathbf{x} \otimes \mathbf{y}) / \|\mathbf{y}\|^2 > 0$ where the first equality holds by the mixed-product property of Kronecker products and positivity holds by [Assumption 1](#). Note that it has negative off-diagonals and so it is an M -matrix and has a nonnegative inverse matrix. Reading off each entry gives:

$$\hat{e}_i^p = \sum_{j=1}^n (\mathbf{M}^{-1})_{ij}(\beta_j^p - r_j \tilde{E}_j) \geq (\mathbf{M}^{-1})_{ii}(\beta_i^p - r_i \tilde{E}_i) > 0$$

where the last inequality holds by [Assumption 2](#). □

A.7 Proof of [Proposition 4](#)

The KKT condition of the principal's problem, which is necessary and sufficient for optimality as all constraints are linear in efforts and the objective is concave, in vector form, is:

$$\nabla_{\mathbf{e}} V(Q^1, \dots, Q^m) + ((\mathbf{G} \otimes \mathbf{I}_m) - (\mathbf{C} \otimes \mathbf{I}_m + \mathbf{R} \otimes (\mathbf{1}_m \mathbf{1}_m^\top - \mathbf{I}_m))) \boldsymbol{\lambda} + \boldsymbol{\mu} = \mathbf{0}$$

where $\boldsymbol{\lambda} \geq \mathbf{0}$ and $\boldsymbol{\mu} \geq \mathbf{0}$ are the vectors of Lagrange multipliers for the incentive constraints and nonnegativity conditions. This is equivalent to $\nabla_{\mathbf{e}} V(Q^1, \dots, Q^m) = \mathbf{W}\boldsymbol{\lambda} - \boldsymbol{\mu}$. If $(i, p) \in A^p$, then $\frac{\partial V}{\partial Q^p} = (\mathbf{W}\boldsymbol{\lambda})_i^p$. If $(i, p) \notin A^p$, then $\frac{\partial V}{\partial Q^p} \leq (\mathbf{W}\boldsymbol{\lambda})_i^p$. Keeping every worker assigned and fully utilized at every project is optimal if and only if $\boldsymbol{\mu} = \mathbf{0}$ and $\nabla_{\mathbf{e}} V(\hat{\mathbf{e}}) = \mathbf{W}\boldsymbol{\lambda}$, which means $\mathbf{W}^{-1} \nabla_{\mathbf{e}} V(\hat{\mathbf{e}}) = \boldsymbol{\lambda} \geq \mathbf{0}$. □

A.8 Proof of [Proposition 5](#)

Let $(Q^{1*}, \dots, Q^{m*})$ be the vector of total efforts that solves the principal's problem. It is unique because the feasible set is compact and the principal's valua-

tion is strictly concave. Now, suppose by contradiction that $Q^{p*} < Q^{q*}$ for some $p < q$. Note that the effort profile that swaps efforts between projects p and q i.e. $Q^{p'} = Q^{q*}$ and $Q^{q'} = Q^{p*}$ is feasible because each worker's incentive constraint remains exactly as it was. Since the valuation is swap-increasing, $(Q^{1'}, \dots, Q^{m'})$ gives a weakly higher value for the principal, which is a contradiction.

Now assume that $p < q$ and $Q^{p*} > Q^{q*}$. Optimality gives $\nabla V(Q^*) \cdot (Q - Q^*) \leq 0$ for every feasible total effort vector Q . Applying this to the swap Q' total effort profile gives $(V^p - V^q)(Q^{q*} - Q^{p*}) \leq 0$. If $Q^{p*} > Q^{q*}$, then it must be that $V^p \geq V^q$. If $Q^{p*} = Q^{q*}$, then the inequality is vacuously true; the inequality $V^p \geq V^q$ must hold because V is swap-increasing. $\square$

A.9 Proof of Proposition 6

I first show that there always exists a worker-symmetric optimal project assignment. Note that the principal's objective is strictly concave over $(Q^1, \dots, Q^m)$. Since the feasible set is convex, there exists a unique maximizer in total efforts per project $(Q^{1*}, \dots, Q^{m*})$. Now, take any optimal project assignment $\{\{\hat{e}_i^p\}_{p=1}^m\}_{i=1}^n$ and consider the average assignment:

$$\bar{e}_i^p := \bar{e}^{p*} = \frac{1}{n} \sum_{j=1}^n \hat{e}_j^p$$

The total efforts at each project remain unchanged, and so $\{\{\bar{e}_i^p\}_{p=1}^m\}_{i=1}^n$ also achieves the principal's maximum. To show it is feasible, note that since $\{\{\hat{e}_i^p\}_{p=1}^m\}_{i=1}^n$ is feasible, then:

$$\beta + g \sum_{j \neq i} \hat{e}_j^p \geq c \hat{e}_i^p + r \sum_{q \neq p} \hat{e}_i^q \implies n\beta + g \sum_{j \neq i} \sum_{\ell=1}^n \hat{e}_\ell^p \geq c \sum_{j=1}^n \hat{e}_j^p + r \sum_{q \neq p} \sum_{j=1}^n \hat{e}_i^q$$

Now we need to show the incentive constraints for $\{\{\bar{e}_i^p\}_{p=1}^m\}_{i=1}^n$ also holds. Take any incentive constraint.

$$\begin{aligned}
\beta + g \sum_{j \neq i} \bar{e}_j^p - c \bar{e}_i^p - r \sum_{q \neq p} \bar{e}_i^q &= \beta + g \sum_{j \neq i} \left(\frac{1}{n} \sum_{\ell=1}^n \hat{e}_\ell^p \right) - c \left(\frac{1}{n} \sum_{j=1}^n \hat{e}_j^p \right) - r \sum_{q \neq p} \left(\frac{1}{n} \sum_{j=1}^n \hat{e}_j^q \right) \\
&= \beta + g \left(\frac{n-1}{n} \sum_{\ell=1}^n \hat{e}_\ell^p \right) - c \left(\frac{1}{n} \sum_{j=1}^n \hat{e}_j^p \right) - r \sum_{q \neq p} \left(\frac{1}{n} \sum_{j=1}^n \hat{e}_j^q \right) \\
&= n\beta + g \left((n-1) \sum_{\ell=1}^n \hat{e}_\ell^p \right) - c \left(\sum_{j=1}^n \hat{e}_j^p \right) - r \sum_{q \neq p} \left(\sum_{j=1}^n \hat{e}_j^q \right) \\
&= n\beta + g \sum_{j \neq i} \sum_{\ell=1}^n \hat{e}_\ell^p - c \sum_{j=1}^n \hat{e}_j^p - r \sum_{q \neq p} \sum_{j=1}^n \hat{e}_j^q \\
&\geq 0
\end{aligned}$$

(1) In the worker-symmetric optimal project assignment, every worker plays the same optimal effort at every project. Because the Cobb-Douglas objective satisfies the Inada conditions, every worker must play positive effort at every project.

(2) The worker-symmetric optimal project assignment solves the symmetric problem:

$$\begin{aligned}
&\max_{\{e^p\}_{p=1}^m} \sum_{p=1}^m \gamma^{(p)} \ln(e^p) \\
&\text{s.t. } \beta + g(n-1)e^p \geq ce^p + r \sum_{q \neq p} e^q
\end{aligned}$$

A unique maximizer $\{e^{p*}\}_{p=1}^m$ exists, and the KKT conditions are both necessary and sufficient. Consider project s and rewrite the incentive constraint in terms of the total effort in the entire system:

$$\beta + g(n-1)e^s \geq ce^s + r \sum_{p \neq s} e^p \implies \beta + (g(n-1) - c + r)e^s \geq r \sum_{p=1}^m e^p$$

For any projects s at which the incentive constraint binds, we have:

$$\beta + (g(n-1) - c + r)e^{s*} = r \sum_{p=1}^m e^{p*} \implies e^{s*} = \frac{\beta - r \sum_{p=1}^m e^{p*}}{c - r - g(n-1)}$$

For project t with a slack incentive constraint, we have

$$\beta + (g(n-1) - c + r)e^{t*} > \sum_{q=1}^m e^{q*} \implies e^{t*} < \frac{\beta - r \sum_{q=1}^m e^{q*}}{c - r - g(n-1)}$$

This means $e^{t*} < e^{s*}$ for any project t and s with slack and binding incentive constraints, respectively.

All projects with binding incentive constraints have the same optimal effort, which depends on the total effort in the entire system. The first-order condition for any project s with a binding incentive constraint is

$$\frac{\gamma^{(s)}}{e^{s*}} = \lambda^{s*}(c - g(n-1) - r) + r \sum_{p=1}^m \lambda^{p*} \geq r \sum_{p=1}^m \lambda^{p*}$$

For project t with a slack incentive constraint ($\lambda^{t*} = 0$), its first-order condition is:

$$\frac{\gamma^{(t)}}{e^{t*}} + \lambda^{t*}(g(n-1) - c + r) - r \sum_{q=1}^m \lambda^{q*} = 0 \implies \frac{\gamma^{(t)}}{e^{t*}} = r \sum_{p=1}^m \lambda^{p*}$$

Linking both inequalities together yields:

$$\gamma^{(s)} \geq e^{s*} \left(r \sum_{p=1}^m \lambda^{p*} \right) > e^{t*} \left(r \sum_{p=1}^m \lambda^{p*} \right) = \gamma^{(t)}$$

Every project with a binding incentive constraint has a strictly larger γ -weight than every project with a slack incentive constraint. Since projects are ordered $\gamma^{(1)} > \dots > \gamma^{(m)}$, that means there exists a project index ϕ^* that for all projects $s \leq \phi^*$, $\gamma^{(s)} \geq \gamma^{(\phi^*)}$ with a binding incentive constraint and for all projects $t > \phi^*$, $\gamma^{(t)} < \gamma^{(\phi^*)}$ with a slack incentive constraint. The formula for ϕ^* is derived in **(4)**.

(3) I first compute the optimal total effort in the entire system, which is a sufficient statistic for efforts at projects with a binding incentive constraint. All projects with binding incentive constraints have the same optimal effort. The following substitutes in the first-order condition for projects with slack incentive constraints.

$$\sum_{p=1}^m e^{p*} = \sum_{p=1}^{\phi^*} e^{p*} + \sum_{t=\phi^*+1}^m e^{t*} = \phi^* e^{s*} + \left(\frac{1}{r \sum_{p=1}^m \lambda^{p*}} \right) \sum_{t=\phi^*+1}^m \gamma^{(t)}$$

From the binding incentive constraint on project s , we obtain:

$$\beta + (g(n-1) - c + r)e^{s*} = r \sum_{p=1}^m e^{p*} \implies \beta = (c - r - g(n-1) + r\phi^*)e^{s*} + \left(\frac{1}{\sum_{p=1}^m \lambda^{p*}} \right) \sum_{t=q^*+1}^m \gamma^{(t)}$$

Next, we determine a relationship between the multipliers $\sum_{p=1}^m \lambda^{p*}$ and the relative project weights γ . Summing the first-order conditions for all projects s with binding incentive constraints gives:

$$\frac{\gamma^{(s)}}{e^{s*}} = (c - r - g(n-1))\lambda^{s*} + r \sum_{p=1}^m \lambda^{p*} \implies \sum_{p=1}^{\phi^*} \frac{\gamma^{(p)}}{e^{p*}} = (c - r - g(n-1)) \sum_{p=1}^{\phi^*} \lambda^{p*} + \phi^* r \sum_{p=1}^m \lambda^{p*}$$

Note that $\sum_{p=1}^{\phi^*} \lambda^{p*} = \sum_{p=1}^m \lambda^{p*}$ because $\lambda^{t*} = 0$ for all $t \geq \phi^*$. Together with the fact that all efforts at projects with binding incentive constraints are identical, we can write:

$$\frac{1}{e^{s*}} \sum_{p=1}^{\phi^*} \gamma^{(p)} = (c - r - g(n-1)) + \phi^* r \sum_{p=1}^m \lambda^{p*} \implies e^{s*}(c - r - g(n-1)) + \phi^* r = \frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{\sum_{p=1}^m \lambda^{p*}}$$

Finally, substitute this into the equation for β :

$$\beta = \frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{\sum_{p=1}^m \lambda^{p*}} + \left(\frac{1}{\sum_{p=1}^m \lambda^{p*}} \right) \sum_{t=\phi^*+1}^m \gamma^{(t)} \implies \sum_{p=1}^m \lambda^{p*} = \frac{\sum_{p=1}^m \gamma^{(p)}}{\beta}$$

Now we can directly compute e^{s*} and e^{t*} :

$$e^{s*} = \frac{\beta - \left(\frac{1}{\sum_{p=1}^m \lambda^{p*}} \right) \sum_{t=\phi^*+1}^m \gamma^{(t)}}{c - r - g(n-1) + r\phi^*} \implies e^{s*} = \frac{\beta}{c - r - g(n-1) + \phi^* r} \left(\frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{\sum_{p=1}^m \gamma^{(p)}} \right)$$

$$e^{t*} = \frac{\gamma^{(t)}}{r \sum_{p=1}^m \lambda^{p*}} = \frac{\beta}{r} \left(\frac{\gamma^{(s)}}{\sum_{p=1}^m \gamma^{(p)}} \right)$$

For project s with a binding incentive constraint, the first-order condition inequal-

ity with the closed form for e^{s^*} gives:

$$\frac{\gamma^{(s)}}{e^{s^*}} \geq r \sum_{p=1}^m \lambda^{p^*} \implies \gamma^{(s)} \left(\frac{c - r - g(n-1) + \phi^* r}{\beta} \right) \left(\frac{\sum_{p=1}^m \gamma^{(p)}}{\sum_{p=1}^{\phi^*} \gamma^{(p)}} \right) \geq r \left(\frac{\sum_{p=1}^m \gamma^{(p)}}{\beta} \right)$$

The right-hand inequality simplifies to

$$\frac{\gamma^{(s)}}{r} \geq \frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{c - r - g(n-1) + \phi^* r}$$

Since this inequality holds for all $s \leq \phi^*$, it must be that

$$\frac{\gamma^{\phi^*}}{r} \geq \frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{c - r - g(n-1) + \phi^* r}$$

Note that project $\phi^* + 1$ must have a slack incentive constraint. We can substitute the formulas for e^{q^*} and e^{q^*+1} into the inequality $e^{\phi^*+1} < e^{q^*}$:

$$\frac{\beta}{r} \left(\frac{\gamma^{(\phi^*+1)}}{\sum_{p=1}^m \gamma^{(p)}} \right) < \frac{\beta}{c - r - g(n-1) + \phi^* r} \left(\frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{\sum_{p=1}^m \gamma^{(p)}} \right) \implies \frac{\gamma^{\phi^*+1}}{r} < \frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{c - r - g(n-1) + \phi^* r}$$

Thus ϕ^* is a natural number that satisfies:

$$\frac{\gamma^{(\phi^*+1)}}{r} < \frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{c - r - g(n-1) + \phi^* r} \leq \frac{\gamma^{(\phi^*)}}{r}$$

If there was another natural number $\hat{q} \neq \phi^*$ that satisfies the above thresholds, then $\hat{q}$ induce $\hat{e}^s \neq e^{s^*}$ and $\hat{e}^t \neq e^{t^*}$ for all projects with binding and slack incentive constraints, respectively. However, this contradicts the uniqueness of the optimal e^* . From the above, we can immediately see that ϕ^* is the maximum project index that satisfies

$$\frac{\sum_{p=1}^{\phi} \gamma^{(p)}}{c - r - g(n-1) + \phi r} \leq \frac{\gamma^{(\phi)}}{r} \iff \frac{\sum_{p=1}^{\phi} \gamma^{(p)}}{\gamma^{(\phi)}} - \phi \leq \frac{c - r - g(n-1)}{r}.$$

(4) I show that ϕ^* is non-increasing in r using the established bounds about ϕ^* .

From the right inequality, we have the equivalent statement

$$\frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{c - r - g(n-1) + \phi^* r} \leq \frac{\gamma^{(\phi^*)}}{r} \iff r \left(\sum_{p=1}^{\phi^*} \gamma^{(p)} - (\phi^* - 1) \gamma^{(\phi^*)} \right) \leq \gamma^{(\phi^*)} (c - g(n-1))$$

Because we ordered $\gamma^{(1)} > \dots > \gamma^{(m)}$,

$$\sum_{p=1}^{\phi^*} \gamma^{(p)} - \phi^* \gamma^{(\phi^*)} > 0 \implies \sum_{p=1}^{\phi^*} \gamma^{(p)} - (\phi^* - 1) \gamma^{(\phi^*)} > 0$$

We derive the following upper bound on r in terms of ϕ^* :

$$r \leq \frac{\gamma^{(\phi^*)} (c - g(n-1))}{\sum_{p=1}^{\phi^*} \gamma^{(p)} - (\phi^* - 1) \gamma^{(\phi^*)}} := U_{\phi^*}$$

From the left inequality that ϕ^* satisfies in (4a), we have the equivalent statement:

$$\frac{\gamma^{(\phi^*+1)}}{r} < \frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{c - r - g(n-1) + \phi^* r} \iff r \left(\sum_{p=1}^{\phi^*} \gamma^{(p)} - (\phi^* - 1) \gamma^{(\phi^*+1)} \right) > \gamma^{(\phi^*+1)} (c - g(n-1))$$

This establishes a lower bound on r in terms of ϕ^* :

$$r > \frac{\gamma^{(\phi^*+1)} (c - g(n-1))}{\sum_{p=1}^{\phi^*} \gamma^{(p)} - (\phi^* - 1) \gamma^{(\phi^*+1)}} := L_{\phi^*}$$

Note that $U_{q+1} = L_q$ and U_q is strictly decreasing in q . I show this by computing the discrete difference

$$U_{q+1} - U_q = \frac{(\gamma^{(q+1)} - \gamma^{(q)}) (c - g(n-1)) \sum_{p=1}^q \gamma^{(p)}}{(\sum_{p=1}^q \gamma^{(p)} - (q-1) \gamma^{(q)}) (\sum_{p=1}^q \gamma^{(p)} - (q-1) \gamma^{(q+1)})} < 0$$

Thus we have partitioned the r -space into m disjoint intervals:

$$[L_m, U_m) \cup [L_{m-1}, U_{m-1}) \cup \dots \cup [L_1, U_1)$$

where $L_m = 0$ and $U_1 = c - g(n-1)$. For any r that satisfies the strict diagonal dominance condition i.e. $r \in [0, c - g(n-1))$, there exists a unique ϕ^* such that

$L_{\phi^*} \leq r < U_{\phi^*}$. Thus ϕ^* is a non-increasing step function of r given by:

$$\phi^*(r) = \max\{q \mid r < U_q\}$$

The bounds U_q are increasing in c and decreasing in g and n . Therefore, ϕ^* is a non-decreasing step function in c and a non-increasing step function in g and n . Because U_q is independent of β , then ϕ^* is also independent of β . $\square$

A.10 Proof of Theorem 2

Rank projects by their marginal value in the optimum and apply Theorem 1 to conclude there exists a worker-specific project index α_i such that $e_i^p > 0$ if and only if $p \leq \phi_i$. To relate α_i to χ_i , I solve for λ_i^p and μ_i^p in the following case analysis. The stationarity equation of the Lagrangian is:

$$V^p + g \sum_{j \neq i} \lambda_j^p - c_i \lambda_i^p - r_i \sum_{q \neq p} \lambda_i^q + \mu_i^p = 0$$

Written in terms of $\Lambda_i := \sum_{p=1}^m \lambda_i^p$ and $\Lambda^p := \sum_{i=1}^n \lambda_i^p$, we have:

$$V^p - \lambda_i^p (c_i + g - r_i) - r_i \Lambda_i + g \Lambda^p + \mu_i^p = 0$$

Case #1.1: $e_i^p > 0$.

Then $\mu_i^p = 0$. From the linear complementarity problem, we have

$$0 = \lambda_i^p (c_i + g - r_i) + r_i \Lambda_i - V^p - g \Lambda^p \implies \lambda_i^p = \frac{V^p + g \Lambda^p - r_i \Lambda_i}{c_i + g - r_i}$$

Since $\lambda_i^p \geq 0$, it must be that $V^p + g \Lambda^p \geq r_i \Lambda_i$.

Case #1.2: $e_i^p = 0$.

Then $\lambda_i^p = 0$ by the generic strict slackness assumption. We have

$$\mu_i^p = r_i \Lambda_i - V^p - g \Lambda^p$$

Putting both cases together, we have the following expressions for λ_i^p and μ_i^p :

$$\lambda_i^p = \frac{\max(V^p + g\Lambda^p - r_i\Lambda_i, 0)}{c_i + g - r_i} \quad \mu_i^p = \max(r_i\Lambda_i - V^p - g\Lambda^p, 0)$$

If $V^p + g\Lambda^p > r_i\Lambda_i$, then $\lambda_i^p > 0$. By [Assumption 3](#), $\lambda_i^p > 0$ implies $e_i^p > 0$. If $V^p + g\Lambda^p < r_i\Lambda_i$, then $\mu_i^p > 0$, forcing $e_i^p = 0$. Finally, if $V^p + g\Lambda^p = r_i\Lambda_i$, then both $\lambda_i^p = \mu_i^p = 0$. This allows us to write $\alpha_i = \max\{p \mid V^p + g\Lambda^p \geq r_i\Lambda_i\}$ and also shows that $V^p + g\Lambda^p$ is non-increasing in p .

Note that the only worker-specific term that determines either multiplier is $r_i\Lambda_i$. I proceed with showing that $\chi_i \geq \chi_j$ implies $r_i\Lambda_i \geq r_j\Lambda_j$. Suppose by contradiction that $r_i\Lambda_i < r_j\Lambda_j$, then $r_j\Lambda_j > 0$ and $\Lambda_j > 0$ necessarily and there is at least one project with $\lambda_j^p > 0$ otherwise $\Lambda_j = 0$. Partition the project space into three sets: $X := \{p \mid V^p + g\Lambda^p > r_j\Lambda_j\}$, $Y := \{p \mid r_j\Lambda_j \geq V^p + g\Lambda^p > r_i\Lambda_i\}$ and $Z := \{p \mid V^p + g\Lambda^p \leq r_i\Lambda_i\}$.

On X , both λ_i^p and λ_j^p are strictly positive and because $r_i\Lambda_i < r_j\Lambda_j$, we have

$$\lambda_i^p = \frac{V^p + g\Lambda^p - r_i\Lambda_i}{c_i + g - r_i} > \frac{V^p + g\Lambda^p - r_j\Lambda_j}{c_i + g - r_i} = \left(\frac{c_j + g - r_j}{c_i + g - r_i} \right) \lambda_j^p$$

On Y , $\lambda_i^p > 0 = \lambda_j^p$ and on Z , both multipliers are zero. Therefore, summing both sides over all projects gives

$$\Lambda_i > \left(\frac{c_j + g - r_j}{c_i + g - r_i} \right) \Lambda_j$$

Multiply both sides by $r_i(c_i + g - r_i)$ and apply $\chi_i \geq \chi_j \iff r_i(c_j + g - r_j) \geq r_j(c_i + g - r_i)$ to obtain

$$r_i(c_i + g - r_i)\Lambda_i > r_i(c_j + g - r_j)\Lambda_j \geq r_j(c_i + g - r_i)\Lambda_j$$

This also holds iff

$$r_i(c_i + g)\Lambda_i > r_j(c_i + g)\Lambda_j$$

This implies $r_i\Lambda_i \geq r_j\Lambda_j$ which is a direct contradiction.

Because $V^p + g\Lambda^p$ is non-increasing in p , the active set for worker i is $\{p \mid V^p +$

$g\Lambda^p \geq r_i\Lambda_i\}$. Thereby $r_i > r_j \implies r_i\Lambda_i \geq r_j\Lambda_j \implies \{p \mid V^p + g\Lambda^p \geq r_i\Lambda_i\} \subseteq \{p \mid V^p + g\Lambda^p \geq r_j\Lambda_j\} \implies \alpha_i \leq \alpha_j$ and $\phi_i \leq \phi_j$. $\square$

A.11 Proof of Theorem 3

The stationarity equation of the Lagrangian is given by

$$V^p + \sum_{j \neq i} g_{ij} \lambda_j^p - c \lambda_i^p - r \sum_{q \neq p} \lambda_i^q + \mu_i^p = 0$$

If every worker is assigned to every project, then $\mu_i^p = 0$ for all (i, p) . Add and subtract $r_i \lambda_i^p$ and stack the stationarity equation for the given project p over all workers and rearrange to obtain:

$$V^p \mathbf{1}_n = \mathbf{R} \Lambda + \mathbf{M} \lambda^p$$

Now sum over all projects:

$$(\sum_{p=1}^m V^p) \mathbf{1}_n = \underbrace{(\mathbf{M} + m\mathbf{R})}_{=\mathbf{P}} \Lambda \implies \Lambda = \mathbf{P}^{-1} (\sum_{p=1}^m V^p) \mathbf{1}_n$$

Substituting into the stationarity condition at project p , we have that

$$V^p \mathbf{1}_n = \mathbf{R} (\mathbf{P}^{-1} (\sum_{p=1}^m V^p) \mathbf{1}_n) + \mathbf{M} \lambda^p \implies \lambda^p = \mathbf{M}^{-1} (V^p \mathbf{1}_n - \mathbf{R} (\mathbf{P}^{-1} (\sum_{p=1}^m V^p) \mathbf{1}_n))$$

Distributing through:

$$\lambda^p = \mathbf{M}^{-1} V^p \mathbf{1}_n - \mathbf{M}^{-1} \mathbf{R} \mathbf{P}^{-1} (\sum_{p=1}^m V^p) \mathbf{1}_n \implies \lambda^p = \mathbf{M}^{-1} V^p \mathbf{1}_n - \frac{1}{m} (\mathbf{M}^{-1} - \mathbf{P}^{-1}) (\sum_{p=1}^m V^p) \mathbf{1}_n$$

The implication uses the following matrix identity

$$\mathbf{M}^{-1} \mathbf{R} \mathbf{P}^{-1} = \frac{1}{m} \mathbf{M}^{-1} (\mathbf{P} - \mathbf{M}) \mathbf{P}^{-1} = \frac{1}{m} (\mathbf{M}^{-1} \mathbf{P} \mathbf{P}^{-1} - \mathbf{M}^{-1} \mathbf{M} \mathbf{P}^{-1}) = \frac{1}{m} (\mathbf{M}^{-1} - \mathbf{P}^{-1})$$

Reading off worker i 's row gives

$$\lambda_i^p = V^p (\mathbf{M}^{-1} \mathbf{1}_n)_i - \frac{1}{m} \sum_{p=1}^m V^p ((\mathbf{M}^{-1} \mathbf{1}_n)_i - (\mathbf{P}^{-1} \mathbf{1}_n)_i)$$

Factoring out $(\mathbf{M}^{-1}\mathbf{1}_n)_i$, we obtain

$$\lambda_i^p = (\mathbf{M}^{-1}\mathbf{1}_n)_i \underbrace{\left(V^p - \frac{1}{m} \sum_{p=1}^m V^p \left(1 - \frac{(\mathbf{P}^{-1}\mathbf{1}_n)_i}{(\mathbf{M}^{-1}\mathbf{1}_n)_i} \right) \right)}_{:=\delta_i}$$

Worker i is fully utilized at project p if and only if $\lambda_i^p > 0$. If $\rho_i > \rho_j$, then $\delta_i < \delta_j$. This implies $\{p \mid V^p > \delta_j\} \subseteq \{p \mid V^p > \delta_i\} \implies \phi_i \geq \phi_j$. $\square$

A.12 Proof of Proposition 7

Since every worker assigned to a project is fully utilized, $\mathbf{W}_\Phi$ is a square matrix. Moreover, it is invertible because it is a principal submatrix of $\mathbf{W}$ which is positive definite. By block matrix inversion, we have

$$\begin{aligned} \mathbf{S} &= (\nabla_{\mathbf{e}}^2 V)^{-1} - (\nabla_{\mathbf{e}}^2 V)^{-1} \mathbf{W}_\Phi (\mathbf{W}_\Phi (\nabla_{\mathbf{e}}^2 V)^{-1} \mathbf{W}_\Phi)^{-1} \mathbf{W}_\Phi (\nabla_{\mathbf{e}}^2 V)^{-1} \\ &= (\nabla_{\mathbf{e}}^2 V)^{-1} - (\nabla_{\mathbf{e}}^2 V)^{-1} \\ &= \mathbf{0} \end{aligned}$$

$\square$

A.13 Proof of Theorem 4

At the optimum, $(\mathbf{e}, \lambda_\Phi)$ satisfy the KKT conditions:

$$\begin{aligned} \nabla_{\mathbf{e}} V(\mathbf{e}) - \mathbf{W}_\Phi^\top \lambda_\Phi &= \mathbf{0} \\ \beta_\Phi - \mathbf{W}_\Phi \mathbf{e} &= \mathbf{0} \end{aligned}$$

Differentiating with respect to parameter θ gives

$$\nabla_{\mathbf{e},\theta}^2 V(\mathbf{e}) + \nabla_{\mathbf{e}}^2 V(\mathbf{e}) \frac{\partial \mathbf{e}}{\partial \theta} - \mathbf{W}_\Phi^\top \frac{\partial \lambda_\Phi}{\partial \theta} - \frac{\partial \mathbf{W}_\Phi^\top}{\partial \theta} \lambda_\Phi = \mathbf{0}$$

$$\frac{\partial \beta_\Phi}{\partial \theta} - \mathbf{W}_\Phi \frac{\partial \mathbf{e}}{\partial \theta} - \frac{\partial \mathbf{W}_\Phi}{\partial \theta} \mathbf{e} = \mathbf{0}$$

Thus we can write in matrix form

$$\underbrace{\begin{bmatrix} \nabla_{\mathbf{e}}^2 V(\mathbf{e}) & -\mathbf{W}_{\Phi}^{\top} \\ -\mathbf{W}_{\Phi} & \mathbf{0} \end{bmatrix}}_{:=\mathbf{J}} \begin{bmatrix} \frac{\partial \mathbf{e}}{\partial \theta} \\ \frac{\partial \lambda_{\Phi}}{\partial \theta} \end{bmatrix} = \begin{bmatrix} -\nabla_{\mathbf{e},\theta}^2 V(\mathbf{e}) + \frac{\partial \mathbf{W}_{\Phi}^{\top}}{\partial \theta} \lambda_{\Phi} \\ \frac{\partial \beta_{\Phi}}{\partial \theta} - \frac{\partial \mathbf{W}_{\Phi}}{\partial \theta} \mathbf{e} \end{bmatrix}$$

Note that $-\nabla_{\mathbf{e},\theta}^2 V(\mathbf{e}) = \mathbf{0}$. I proceed to show that $\mathbf{J}$ is invertible. Suppose $\mathbf{x} \in \mathbb{R}^{mn}$ and $\mathbf{y} \in \mathbb{R}^F$ where $(\mathbf{x}, \mathbf{y}) \in \ker(\mathbf{J})$. Therefore:

$$\begin{aligned} \nabla_{\mathbf{e}}^2 V(\mathbf{e})\mathbf{x} - \mathbf{W}_F^{\top} \mathbf{y} &= \mathbf{0} \\ -\mathbf{W}_F \mathbf{x} &= \mathbf{0} \end{aligned}$$

Premultiplying the first equation by $\mathbf{x}^{\top}$ gives

$$\mathbf{x}^{\top} \nabla_{\mathbf{e}}^2 V(\mathbf{e})\mathbf{x} - \mathbf{x}^{\top} \mathbf{W}_F^{\top} \mathbf{y} = 0 \implies \mathbf{x}^{\top} \nabla_{\mathbf{e}}^2 V(\mathbf{e})\mathbf{x} = 0$$

By assumption that $\nabla_{\mathbf{e}}^2 V(\mathbf{e})$ is strictly negative definite on $\ker(\mathbf{W}_{\Phi}^{\top})$, it must be that $\mathbf{x} = \mathbf{0}$. Substituting back into the first equation, we get

$$\mathbf{W}_{\Phi}^{\top} \mathbf{y} = 0$$

Recall that $\mathbf{W}$ is invertible, so $\mathbf{W}_{\Phi}$ has full row rank and it must be that $\mathbf{y} = \mathbf{0}$. This concludes the argument that $\mathbf{J}$ is invertible. Therefore the Implicit Function Theorem applies, and

$$\begin{bmatrix} \frac{\partial \mathbf{e}}{\partial \theta} \\ \frac{\partial \lambda_{\Phi}}{\partial \theta} \end{bmatrix} = \mathbf{J}^{-1} \begin{bmatrix} \frac{\partial \mathbf{W}_{\Phi}^{\top}}{\partial \theta} \lambda_{\Phi} \\ \frac{\partial \beta_{\Phi}}{\partial \theta} - \frac{\partial \mathbf{W}_{\Phi}}{\partial \theta} \mathbf{e} \end{bmatrix}$$

which implies that

$$\frac{\partial \mathbf{e}}{\partial \theta} = \mathbf{S}(\frac{\partial \mathbf{W}_{\Phi}^{\top}}{\partial \theta} \lambda_{\Phi}) + \mathbf{Y}(\frac{\partial \beta_{\Phi}}{\partial \theta} - \frac{\partial \mathbf{W}_{\Phi}}{\partial \theta} \mathbf{e})$$

□

A.14 Proof of Proposition 8

Under full utilization and activity, the optimal project assignment has, for each worker i and project p , $e_i^{p*} > 0$, and the incentive constraint binds

$$\beta_i^p + \sum_{j \neq i} g_{ji} e_j^{p*} = c_i e_i^{p*} + r_i \sum_{q \neq p} e_i^{q*}$$

Therefore, we have

$$\beta^p = \mathbf{M} \mathbf{e}^p + \mathbf{R} \mathbf{E} \mathbf{1}_m$$

Now sum over all projects to obtain

$$\sum_{p=1}^m \beta^p = \sum_{p=1}^m (\mathbf{M} \mathbf{e}^p + \mathbf{R} \mathbf{E} \mathbf{1}_m) \implies \sum_{p=1}^m \beta^p = \mathbf{M} \mathbf{E} \mathbf{1}_m + m \mathbf{R} \mathbf{E} \mathbf{1}_m$$

Therefore the total effort vector satisfies $\mathbf{E} \mathbf{1}_m = \mathbf{P}^{-1} \sum_{p=1}^m \beta^p$. Substituting into the incentive constraint for project p gives

$$\beta^p = \mathbf{M} \mathbf{e}^p + \mathbf{R} \mathbf{P}^{-1} \sum_{p=1}^m \beta^p \mathbf{1}_m \implies \mathbf{e}^p = \mathbf{M}^{-1} \beta^p - \mathbf{M}^{-1} \mathbf{R} \mathbf{P}^{-1} \sum_{p=1}^m \beta^p$$

Now we can differentiate

$$\frac{\partial \mathbf{e}^p}{\partial \beta_j^q} = \mathbf{M}^{-1} \epsilon_j \delta_{pq} - \mathbf{M}^{-1} \mathbf{R} \mathbf{P}^{-1} \epsilon_j$$

Reading off the i -th entry gives the formula in the theorem. All of the other equations are derived similarly.

To show the sign of $\frac{\partial e_i^{p*}}{\partial \beta_j^q}$, note that its matrix form is given by:

$$\frac{\partial \mathbf{e}^p}{\partial \beta^q} = \mathbf{M}^{-1} \delta_{pq} - \mathbf{M}^{-1} \mathbf{R} \mathbf{P}^{-1} = \mathbf{M}^{-1} \delta_{pq} - \frac{1}{m} (\mathbf{M}^{-1} - \mathbf{P}^{-1})$$

The second equality hold because $\mathbf{R} = \frac{1}{m}(\mathbf{P} - \mathbf{M})$. Therefore:

$$\frac{\partial \mathbf{e}^p}{\partial \beta^q} = \begin{cases} \frac{m-1}{m}\mathbf{M}^{-1} + \frac{1}{m}\mathbf{P}^{-1} & p = q \\ -\frac{1}{m}(\mathbf{M}^{-1} - \mathbf{P}^{-1}) & p \neq q \end{cases}$$

Therefore $\frac{\partial e_i^{p*}}{\partial \beta_j^p} > 0$ for all $j \neq i$ and $\frac{\partial e_i^{p*}}{\partial \beta_j^q} < 0$ for all $q \neq p$. $\square$

A.15 Proof of Proposition 9

As an abuse of notation, $\mathbf{e}^p$ and β^p both refer to the efforts and preferences of workers restricted to those in A^p . Define $\mathbf{E} = [\mathbf{e}^1 \cdots \mathbf{e}^n]$. In an optimum with full utilization and partial activity, the binding incentive constraint for worker i at project p is

$$\beta_i^p + \sum_{j \neq i, j \in A^p} g_{ji} e_j^{p*} = c_i e_i^{p*} + r_i \sum_{q \neq p, q \in P_i} e_i^{q*}$$

In vector form, we have

$$\beta^p = \mathbf{M}^p \mathbf{e}^p + (\mathbf{R} \mathbf{E} \mathbf{1}_m)^p$$

where $(\mathbf{R} \mathbf{E} \mathbf{1}_m)^p$ is the vector of total efforts for workers in A^p multiplied by r_i . Solving for $\mathbf{e}^p$ gives

$$\mathbf{e}^p = (\mathbf{M}^p)^{-1} \beta^p - (\mathbf{M}^p)^{-1} (\mathbf{R} \mathbf{E} \mathbf{1}_m)^p$$

$\mathbf{M}^p$ is invertible because principal submatrices of positive definite matrices are also positive definite by Sylvester's Criterion. Now I solve for a worker's total effort

$$Q_i = \sum_{p \in P_i} e_i^p = \sum_{p \in P_i} \left(\sum_{j \in A^p} (\mathbf{M}^p)^{-1}_{ij} \beta_j^p - (\mathbf{M}^p)^{-1}_{ij} r_j Q_j \right)$$

Rearranging, I obtain:

$$Q_i + \sum_{j=1}^n \left(r_j \sum_{p \in P_i \cap P_j} (\mathbf{M}^p)^{-1}_{ij} \right) Q_j = \sum_{p \in P_i} \sum_{j \in A^p} (\mathbf{M}^p)^{-1}_{ij} \beta_j^p$$

Define $\hat{\beta}_i := \sum_{p \in P_i} \sum_{j \in A^p} (\mathbf{M}^p)_{ij}^{-1} \beta_j^p$ and $n \times n$ matrix $\mathcal{P}$ with the following entries:

$$\mathcal{P}_{ij} = \delta_{ij} + r_j \sum_{p \in P_i \cap P_j} (\mathbf{M}^p)_{ij}^{-1}$$

Given $\mathcal{P}$, we have the following linear system in workers' total efforts

$$\mathcal{P}(\mathbf{E}\mathbf{1}_m) = \hat{\beta}$$

Because V is strictly concave in total efforts, the optimal total effort is unique and $\mathcal{P}$ is invertible.

$$\mathbf{E}\mathbf{1}_m = \mathcal{P}^{-1} \hat{\beta} \implies \frac{\partial(\mathbf{E}\mathbf{1}_m)}{\partial \zeta} = \mathcal{P}^{-1} \boldsymbol{\zeta}^{(\zeta)}$$

where $\boldsymbol{\zeta}^{(\zeta)}$ is defined for each model parameter ζ as

$$\boldsymbol{\zeta}_i^{(\beta_j^q)} = \begin{cases} (\mathbf{M}^q)_{ij}^{-1} & \text{if } i \in A^q \\ 0 & \text{o.w.} \end{cases} ; \boldsymbol{\zeta}_i^{(c_j)} = - \sum_{p \in A_i \cap A_j} (\mathbf{M}^p)_{ij}^{-1} e_j^{p*}$$

$$\boldsymbol{\zeta}_i^{(r_j)} = \sum_{p \in A_i \cap A_j} (\mathbf{M}^p)_{ij}^{-1} (e_j^{p*} - Q_j^*) ; \boldsymbol{\zeta}_i^{(g_{j\ell})} = \sum_{p \in A_i \cap A_j \cap A_\ell} (\mathbf{M}^p)_{ij}^{-1} e_\ell^{p*} + (\mathbf{M}^p)_{i\ell}^{-1} e_j^{p*}$$

Now we can finally compute:

$$\frac{\partial \mathbf{e}^p}{\partial \zeta} = \frac{\partial}{\partial \zeta} \left((\mathbf{M}^p)^{-1} \boldsymbol{\beta}^p \right) - \frac{\partial}{\partial \zeta} \left((\mathbf{M}^p)^{-1} (\mathbf{R}\mathbf{E}\mathbf{1}_m)^p \right)$$

Each of the results follows from direct computation of the above equation and substituting in $\frac{\partial(\mathbf{E}\mathbf{1}_m)}{\partial \zeta} = \mathcal{P}^{-1} \boldsymbol{\zeta}^{(\zeta)}$. $\square$

A.16 Proof of Proposition 10

Assume that $\mathbf{e}^*$ is an effort profile where worker i is underutilized at p so that $\frac{\partial U_i}{\partial e_i^p} > 0$ at $\mathbf{e}^*$. Since U_i is continuously differentiable, there exists $\epsilon > 0$ where $U_i(\mathbf{e}_i^* + \mathbf{1}_m \cdot \epsilon, \mathbf{e}_{-i}^*) > U_i(\mathbf{e}_i^*, \mathbf{e}_{-i}^*)$. Monotonicity of w_i gives $w_i(\mathbf{e}_i^* + \mathbf{1}_m \cdot \epsilon, \mathbf{e}_{-i}^*) \geq w_i(\mathbf{e}_i^*, \mathbf{e}_{-i}^*)$, and so we have constructed a profitable deviation for worker i . $\square$

B The General MINLP

The principal's general problem is written as a mixed-integer nonlinear program (MINLP).

$$\begin{aligned}
& \max_{\mathbf{e}, \mathbf{z}} V(Q^1, \dots, Q^m) \\
& \text{s.t. } \mathbf{e} \geq \mathbf{0}, \\
& \mathbf{z} \in \{0, 1\}^{mn}, \\
& \forall(i, p), e_i^p \leq Mz_i^p \\
& \forall(i, p), \beta_i^p + \sum_{j \neq i} g_{ij}e_j^p - c_i e_i^p - r_i \sum_{q \neq p} e_i^q \geq -M(1 - z_i^p) \\
& \forall p, \sum_{i=1}^n e_i^p = Q^p
\end{aligned} \tag{P3}$$

where M is a sufficiently large positive constant.²⁰ z_i^p is a binary variable that acts as a switch to ensure that if $e_i^p > 0$, then the incentive constraint has to be satisfied. If that is the case, then z_i^p must be 1. Otherwise, if $e_i^p = 0$, then no incentive constraint is needed and z_i^p is set to 0. Switching the incentive constraints turns the principal's problem into a mixed-integer nonlinear programming problem, rendering it *NP*-hard.

To see an explicit example of parameters where the feasible set is nonconvex, take a simple example with one worker i and two projects, p and q . Let the productivity parameters be $\beta_i^p = 4$, $\beta_i^q = 1$, $c_i = 0.5$, $r_i = 0.3$. He prefers working on project p four times as much as on project q . The feasible set contains all pair of nonnegative efforts (e_i^p, e_i^q) that satisfy:

$$\begin{aligned}
4 &\geq 0.5e_i^p + 0.3e_i^q & \text{if } e_i^p > 0 \\
1 &\geq 0.5e_i^q + 0.3e_i^p & \text{if } e_i^q > 0
\end{aligned}$$

Figure 8 illustrates the boundary of the set of implementable efforts. Effort vectors

²⁰For example, one can set $M = \|(\mathbf{C} \otimes \mathbf{I}_m - \mathbf{G} \otimes \mathbf{I}_m)^{-1} \boldsymbol{\beta}\|_\infty$. $(\mathbf{C} \otimes \mathbf{I}_m - \mathbf{G} \otimes \mathbf{I}_m)^{-1} \boldsymbol{\beta}$ gives the vector of maximum efforts workers could play at each project, absent any substitutability across projects. Instead of using a Big- M constant, we can also write the principal's problem with disjunctive constraints: if $z_i^p = 1$, then $\beta_i^p + \sum_{j \neq i} g_{ij}e_j^p - c_i e_i^p - r_i \sum_{q \neq p} e_i^q \geq 0$; else $e_i^p = 0$.

of the form $(e_i^p, 0)$ for $e_i^p \in [3\frac{1}{3}, 4]$ are feasible, as $0.5e_i^p \leq 2$. e_i^q is zero at those efforts, and so his incentive constraint for project q has no bite.

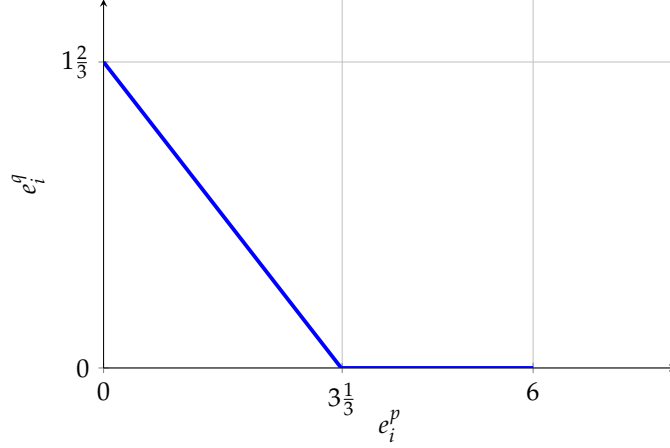


Figure 8: Nonconvex Boundary of Implementable Efforts

The inherent difficulty in analyzing the MINLP comes from the combinatorial nature of the problem. Traditional approaches such as using a greedy algorithm fail because it is neither monotone nor submodular. Increasing the effort cap for a worker at a particular project may cause him to reduce efforts at other projects, leading to an overall decrease in the principal's valuation. Raising effort caps could also cause other workers to increase their efforts on that particular project due to complementarity, increasing the principal's valuation in a convex fashion.

Algorithmic methods, such as outer approximations, serve as efficient ways to compute the solution to the MINLP. Unfortunately, algorithmic approaches provide little in terms of structural and economic insights. Alternatively, absent the sufficient condition on the model parameters in [Proposition 2](#), we can interpret the comparative statics exercises as local analysis treating the binary $\mathbf{z}$ variables in the MINLP as fixed.

C Generic Conditions in the Optimum

Statement (1) is the usual strict complementary slackness condition on λ , which is generic for convex optimization problems that satisfy strict feasibility.

To show that statement (2), $e_i^p = 0$ implies $\lambda_i^p = 0$, is a generic condition, I argue that the set of model parameters that satisfy [Assumption 3](#) with $e_i^p = 0$ and a binding incentive constraint for some worker-project pair (i, p) has measure zero. I label this set as $\mathcal{M}$. Crucially, this *does not* imply the converse i.e. that $\lambda_i^p = 0$ implies $e_i^p = 0$. In the environment with homogeneous workers, underutilizing workers, i.e., $\lambda_i^p = 0$ with $e_i^p > 0$, is a generic feature of the optimal project assignment.

I write $(\beta, \mathbf{c}, \mathbf{g}, \mathbf{r})$ to be an instance of the model parameters. First, note that the planner's objective is strictly concave in total efforts, the feasible set is convex, Slater's condition holds, and the optimal effort totals $Q^p(\beta, \mathbf{c}, \mathbf{g}, \mathbf{r})$ and $Q_i(\beta, \mathbf{c}, \mathbf{g}, \mathbf{r})$ are unique. On an open, dense subset $\mathcal{U}$ of the space of valid model parameters, the Implicit Function Theorem implies that $Q^p(\beta, \mathbf{c}, \mathbf{g}, \mathbf{r})$ and $Q_i(\beta, \mathbf{c}, \mathbf{g}, \mathbf{r})$ are real-analytic functions. Assume by contradiction that $\mathcal{M}$ has a non-empty interior. This implies the subset $\mathcal{M}_i^p(\beta, \mathbf{c}, \mathbf{g}, \mathbf{r}) := \beta_i^p + gQ^p(\beta, \mathbf{c}, \mathbf{g}, \mathbf{r}) - r_i Q_i(\beta, \mathbf{c}, \mathbf{g}, \mathbf{r})$ vanishes on an open set within $\mathcal{U}$ for some (i, p) . By the Identity Theorem for real-analytic functions, $\mathcal{M}_i^p(\beta, \mathbf{c}, \mathbf{g}, \mathbf{r}) \equiv 0$ everywhere on $\mathcal{U}$.

By continuity, there exists some $\epsilon > 0$ sufficiently small such that $r_j = \epsilon$ for all workers implies every worker is fully active and fully utilized at every project. With vanishing cross-project costs, each worker treats each project independently. Since V is strictly increasing in each argument, the principal assigns each worker to each project and pushes them to exert the highest feasible effort, made possible by fully utilizing them. Thus, $\mathcal{M}_i^p(\beta, \mathbf{c}, \mathbf{g}, \mathbf{r}) > 0$ for a set of model parameters $(\beta, \mathbf{c}, \mathbf{g}, \mathbf{r})$ with positive measure. Thus the set of zeros of $\mathcal{M}_i^p$ cannot contain an open set and must be measure zero. Thereby, the union of the set of zeros $\mathcal{M} = \cup_{i,p} \mathcal{M}_i^p$ also has measure zero. We conclude that $e_i^p = 0$ implies $\lambda_i^p = 0$ is a generic property of the solution and can be assumed without loss of generality.

D An Endogenous Sorting Index

Why is it that once the principal leaves workers off some projects in the optimum, the ordering of the cutoffs cannot be read off the primitives? Endogeneity resides in the numerator of the scope index. In general, it is worker i 's total shadow value $\Lambda_i = \sum_{p=1}^m \lambda_i^p$ where λ_i^p is the multiplier on worker i 's incentive constraint at

project p . It measures what the principal gains if she could relax worker i 's incentive constraints at every project; generalists have higher Λ_i . Under full assignment, the total shadow value is proportional to a worker i 's Katz-Bonacich centrality in $\mathbf{P}$. However, this is not true when a worker is assigned to fewer projects in the optimum. His total shadow value depends on his marginal cost of total effort, which is no longer $c_i + r_i(m - 1)$. Instead, it depends on the number of projects he is assigned to, which is endogenous. Thus, Λ_i is not related to $\mathbf{P}$ and depends on the assignment pattern itself.

In general, assignment and full utilization cutoffs are ordered by an endogenous index I call the *congestion index*. It is formulated recursively: once a worker is dropped from a team, he stops generating complementarities at that project and the network on which the index is defined shrinks as we descend the project ranking.

Definition 10. For a subset of workers $S \subseteq N$, the congestion index of worker j on S is:

$$\eta_j(S) := \frac{r((\mathbf{M}^S)^{-1}\mathbf{\Lambda}^S)_j}{((\mathbf{M}^S)^{-1}\mathbf{1}^S)_j}$$

where $\mathbf{M}^S$, $\mathbf{\Lambda}^S$ and $\mathbf{1}^S$ are the submatrices and subvectors with indices restricted to those in S and $\mathbf{\Lambda} := [\sum_{p=1}^m \lambda_1^p \cdots \sum_{p=1}^m \lambda_n^p]^\top$ where λ_i^p is the multiplier on worker i 's incentive constraint at project p in the optimum. The ordering of workers is recursive. Set $N_0 := N$ and for $k = 1, \dots, n$, compute

$$i(k) := \arg \max_{j \in N_{k-1}} \eta_j(N_{k-1}) \text{ where } N_k := N_{k-1} \setminus \{i(k)\}$$

Relabel worker indices in order $i(1), \dots, i(n)$. Worker i 's **congestion index** is given by $\eta_i := \eta_i(N_{i-1})$.

The congestion index η_i is an endogenous ratio of two centralities which captures the trade-off. The denominator is the canonical Katz-Bonacich (KB) centrality $((\mathbf{M}^S)^{-1}\mathbf{1})_i$ with the team interaction matrix restricted to members in S . It counts the total additional effort that comes from relaxing worker i 's incentive constraint at a project, incorporating both the direct effect for worker i and all of the complementarity effects for other workers. The numerator applies the same team interaction matrix to Λ_i , which is the total shadow value of worker i 's incentive con-

straints. It measures the opportunity cost incurred by the principal for all other projects when she increases worker i 's effort towards a specific project, scaled by r . A high congestion index means it is better to concentrate that worker on only a few projects.

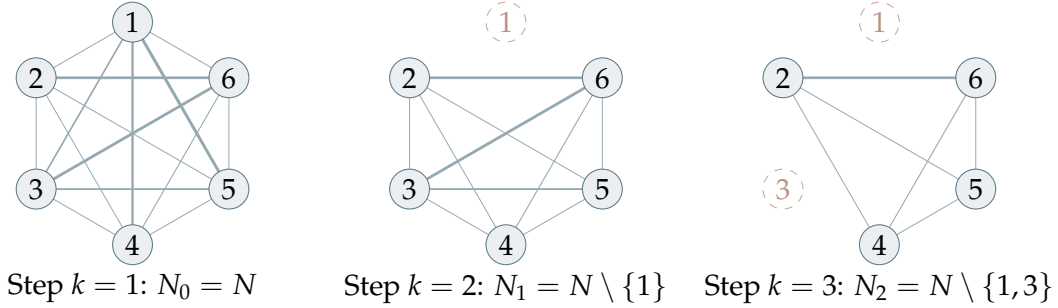


Figure 9: Once a worker is removed from a project, he stops generating complementarity to those who remain, so $\eta_i(\cdot)$ is recomputed on a strictly smaller network.

Once a specialist is identified and left off projects, he is no longer a source of complementarity for any worker left at those projects. The congestion index is recomputed for the remaining workers on the submatrix of $\mathbf{M}$ *leaving out* worker i . This result is descriptive rather than prescriptive and I emphasize that sorting cannot be achieved through just the primitives because the multipliers Λ cannot be solved in closed form.

Theorem 5. *The assignment and full utilization cutoffs α_i and ϕ_i are sorted such that $\eta_i > \eta_j$ if and only if $\alpha_i \leq \alpha_j$ and $\phi_i \leq \phi_j$.*

Proof. The linear complementarity problem that the multipliers solve is given by

$$\mu^p = \mathbf{M}\lambda^p + \mathbf{R}\Lambda - V^p \mathbf{1}_n; \lambda^p, \mu^p \geq \mathbf{0}; (\lambda^p)^\top \mu^p = 0$$

To understand when workers are unassigned from projects due to a decreasing marginal value of projects V^p , which is a discrete parameter, I turn to its continuous counterpart.

$$\mu^p = \mathbf{M}\lambda^p + \mathbf{R}\Lambda - \eta \mathbf{1}_n; \lambda^p, \mu^p \geq \mathbf{0}; (\lambda^p)^\top \mu^p = 0$$

V^p is replaced by $\eta \in \mathbb{R}$. I employ a pivot-type method. Ignoring non-negativity of λ^p , I solve the system assuming that $\mu^p = \mathbf{0}$, i.e., all workers are active on project

p . We obtain:

$$\mathbf{M}\lambda^p + \mathbf{R}\Lambda - \eta \mathbf{1}_n = \mathbf{0} \implies \lambda^p = \mathbf{M}^{-1}(\eta \mathbf{1}_n - \mathbf{R}\Lambda)$$

If η is sufficiently large, then $\lambda^p > \mathbf{0}$ and all workers are assigned positive effort on project p . As η decreases, λ^p likewise decreases. The first index i of λ^p to reach zero is when

$$\eta = \left(\frac{r(\mathbf{M}^{-1}\Lambda)_i}{(\mathbf{M}^{-1}\mathbf{1})_i} \right)$$

and it becomes negative if

$$\eta < \left(\frac{r(\mathbf{M}^{-1}\Lambda)_i}{(\mathbf{M}^{-1}\mathbf{1})_i} \right)$$

Thus the index i corresponding to $\arg \max_i \left(\frac{r(\mathbf{M}^{-1}\Lambda)_i}{(\mathbf{M}^{-1}\mathbf{1})_i} \right)$ is the first worker to be assigned $\mu_i^p > 0$, else $\lambda_i^p < 0$ which contradicts the latter's nonnegativity. This implies worker i must be unassigned from all projects p with

$$V^p < \left(\frac{r(\mathbf{M}^{-1}\Lambda)_i}{(\mathbf{M}^{-1}\mathbf{1})_i} \right)$$

On those projects, we can employ the same strategy, but now restrict attention to the first index j of $\lambda^{p, N \setminus \{i\}}$ to reach zero, which is when

$$\eta = \left(\frac{r((\mathbf{M}^{N \setminus \{i\}})^{-1}\Lambda^{N \setminus \{i\}})_i}{((\mathbf{M}^{N \setminus \{i\}})^{-1}\mathbf{1}^{N \setminus \{i\}})_j} \right)$$

This is precisely because worker i is no longer an active participant on those projects. The same logic extends recursively to all remaining workers. $\square$